A
Project Report
on

Agro Waste

BTech-IT, Sem VI

Prepared By:
Jenish Patel (IT105)
Deep Patel (IT094)
Het Patel (IT104)

Guided By:
Prof. Deepak C Vegda

Dept. of
Information Technology

Department of Information Technology
Faculty of technology,
Dharmsinh Desai University
College road, Nadiad- 387001

April, 2023

CANDIDATE'S DECLARATION

We declare that the pre-final semester report entitled "**Agro Waste**" is our own work conducted under the supervision of the guide **Prof. Deepak C Vegda.**

We further declare that to the best of our knowledge the report for B.Tech. VI semester does not contain part of the work which has been submitted either in this or any other university without proper citation.

Candidate's Signature
Jenish Patel
Student ID: 20ITUBS039

Candidate's Signature
Deep Patel
Student ID: 20ITUBS141

Candidate's Signature
Het Patel
Student ID: 20ITUOS130

DHARMSINH DESAI UNIVERSITY

NADIAD-387001, GUJARAT

CERTIFICATE

This is to certify that the project carried out in the subject of System Design Practice, entitled "Agro Waste "and recorded in this reportis a Bonafide report of work of

1) Jenish Niteshkumar Patel	**Roll No. IT105**	**ID No.: 20ITUBS039**
2) Deep Bhikhubhai Patel	**Roll No. IT094**	**ID No: 20ITUBS141**
3) Het Rakeshbhai Patel	**Roll No. IT104**	**ID No: 20ITUOS130**

of Department of Information Technology, semester VI. They were involved in Project work during the academic year 2022 -2023.

Prof. Deepak C Vegda
(Project Guide),
Department of Information Technology,
Faculty of Technology,
Dharmsinh Desai University, Nadiad
Date:

ACKNOWLEDGEMENT

It is indeed a great pleasure to express our thanks and gratitude to all those who helped us during this project. This project has given us a great opportunity to think, implement and interact with various aspects of the Software Development Life Cycle. We would like to acknowledge all the people who have helped us at one stage or another by providing the much-needed support, encouragement and groundwork to complete our project.

We express a deep sense of gratitude towards our project guide **Prof. Deepak C Vegda** towards his innovative ideas and earnest effort to make our project a success. It is his sincerity that prompted us throughout the project to do hard work using industry adopted technologies. Our commitment to the application is the sole result of patience, hard work and dedication being inspired by him.

A blend of gratitude, pleasure and great satisfaction is what we feel to convey our indebtedness to all those who all have directly or indirectly contributed towards completion of the project.

with sincere regards,
Deep Patel,
Het Patel,
Jenish Patel

TABLE OF CONTENTS

ABSTRACT

This project submission proposes an Agro-Waste Management system. As we know residue burning is a major problem in Northern part of India and also Government of Delhi-Punjab using millions of amounts from their budget to face the this issue. Our project proposes a website in which Farmers of Punjab has to register themself with crop details, Land coverage details at the time of sowing. At the moment of harvesting farmers can request for this harvesting and residue was assembled and collected by Government and stored it to warehouses. Revenue of the this collected residue will be given to the farmers as per standard rates.

Analysis of warehousing, imports and export will give ideas towards what to do next! based on this analysis Government has to held Auction and declare notification of invitation via social network. Companies and Industries can Register and place-bids on required products. Result will be generated in favor of highest placed-bid after deadline.

Supervised Image Classification (SIC) is also used using GIS(Geographic Information System(QGIS(Quantum Geographic Information System))).In this using Semi-Automatic Classification Plugin (SCP) all the training inputs will be stored into .scp file. Using this file analysis and classification of satellite imagery(tif) of any part will be easy.

Overall, objective of this project is to well-planned management of crop residue and enhance our Northern-India environment. Also SIC gives idea to the Government that How much expected residue will be generated and How much efforts are required. Hence, Analysis and preplanning will be easy.

LIST OF FIGURES

LIST OF TABLES

1. INTRODUCTION

1.1. Purpose

This document aims to build an online website platform to ease the process of residue management, coordination and communication among placement farmers, companies and Government. The system will be based on a web application that the farmers can use to request for service or notify Government of new service openings. The system will have a simple UX and will be secured through the encryption of important data and will be fast enough to provide access to stored information in a reasonable amount of time.

1.2. Document Conventions

This document follows MLA format. Bold-faced text has been used to emphasize section and sub-section headings. Italicized text is used to label and recognize diagrams.

1.3. Architecture Diagram

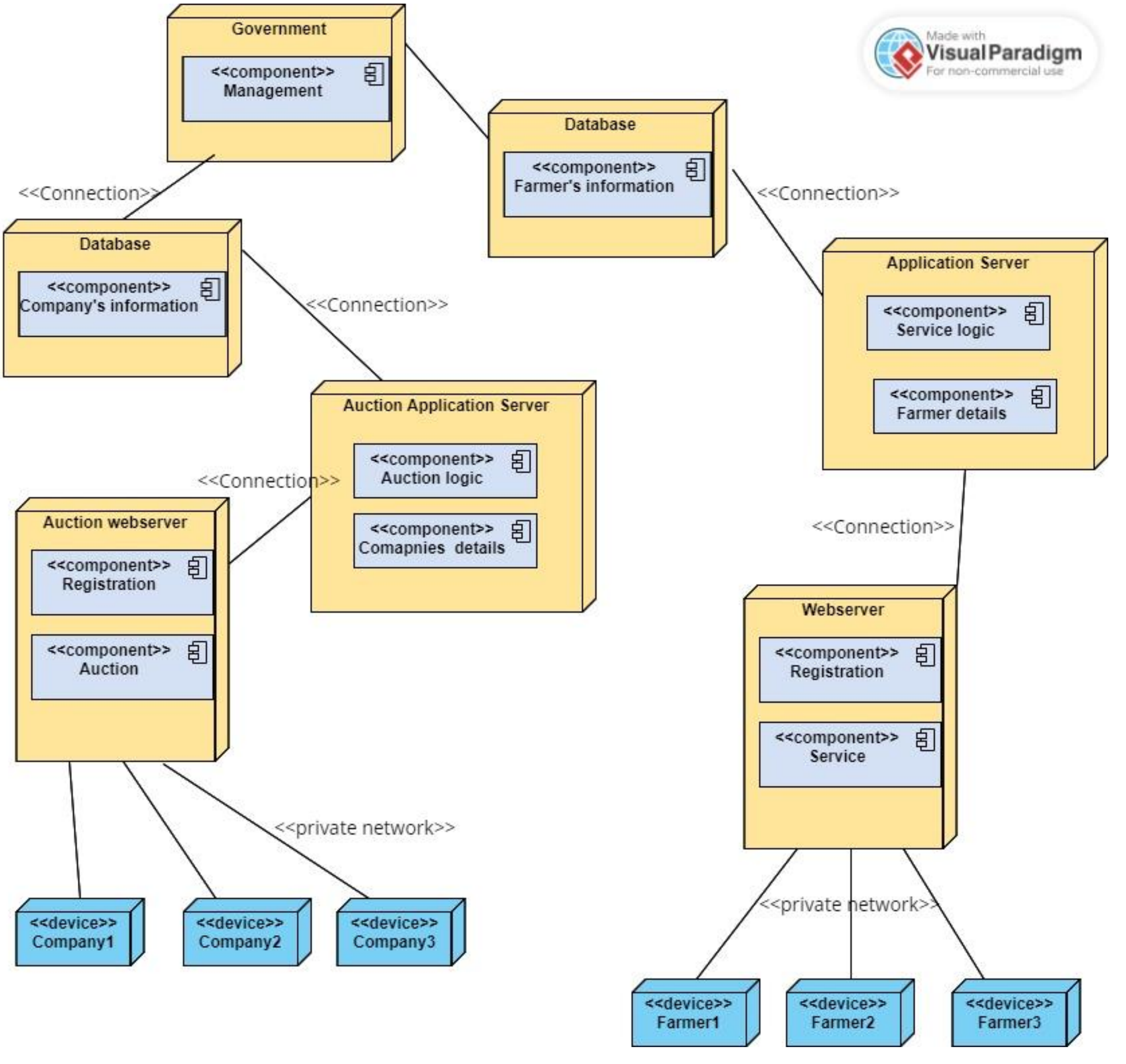


Fig 1.1
(Architecture Diagram)

1.4. Intended Audience and Reading Suggestions

- Developers
- Project Managers
- Marketing staff
- Users
- Testers
- Documentation Writers

1.5. Product Scope

The purpose of this project is to provide ease for the process of Crop residue management. Farmers can notify about service to Government via website. Farmers has to upload their details on the platform at the time of sowing. Government can conduct an auction for residue in which respected Companies can place bid via website.

1.6. Technology and Literature Review

Following technologies will be used for development/management/tracking activities….

1) Visual Studio Code for developing web applications.

2) Mongo Database System for Data Storage and Management thus providing 24*7 service.

2. PROJECT MANAGEMENT

2.1. Feasibility Study

2.1.1. Technical feasibility

Viewing our project from a technical point of view (thinking about various tools and technologies being used in developing the system). We have decided that the following technologies will be morethan enough to develop a complete working system (including tech &tools used for project tracking, monitoring etc. along with development).

For web app development: VS Code

Database: Mongo database

We are equipped with basic workflows of each tool and tech. and capable of exploring further if required. Each of the above technologies is freely available and some of the skills yet to be learnt are manageable. From this, our project is technically feasible.

2.1.2. Time schedule feasibility

We have planned the steps for completion of our project in the given duration. Firstly, we will perform requirement gathering & analysis by the start of January 2023. We will prepare SRS document andthe GUI design tentatively by January 2023 ending. The diagrams required for the design as well as the database design will be tentatively completed by February 2023.For coding and unit testing 4weeks and for system and integration testing another 2 weeks will be required. Hence tentatively by the end of March 2023. We will be ableto complete the project and ready for its demonstration at the start of April 2023. Being a 3-member team, we will be able to complete our project in the estimated time.

2.1.3. Economic feasibility

This evaluation often includes a cost/benefit analysis of the project, which assists businesses in determining the viability, cost, and advantages of a project before allocating financial resources.

It also functions as an impartial project evaluation and enhances project credibility by assisting decision-makers in determining the positive economic advantages that the proposed project would give to the business.

.

2.1.4. Implementation feasibility

We will be working on developing a full Web application for the first time. So, we need to learn the basics of MERN stack. Also, we needed to learn how to connect our project with an online database MongoDB.

Since we are aware of the basics of JavaScript, we just needto learn how to implement it according to our needs which will take around 2 or 3 weeks and be completed before starting implementation.

2.2. Project Planning

2.2.1. Project Development Approach and Justification

We will be using the Agile model for project development. Agile methods break tasks into smaller iterations, or parts do not directly involve long term planning. The project scope and requirements are laid down at the beginning of the development process. Plans regarding the number of iterations, the duration and the scope of each iteration are clearly defined in advance.

Following are the phases in the Agile model are as follows:

1. Requirements gathering

2. Design the requirements

3. Construction/ iteration

4. Testing/ Quality assurance

5. Deployment

6. Feedback

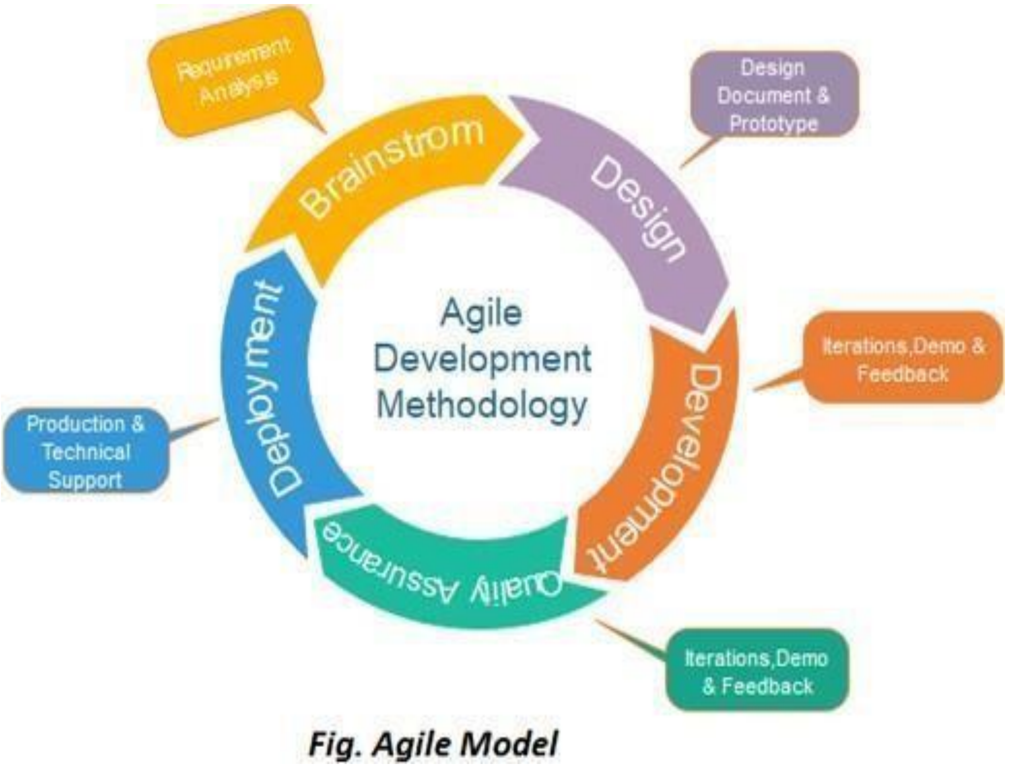


Fig 2.2.1 Agile Model

Advantages of Agile model:

- Customer satisfaction by rapid, continuous delivery of useful software.
- People and interactions are emphasized rather than processes and tools.Customers, developers, and testers constantly interact with each other.
- Working software is delivered frequently (weeks rather than months).
- Face-to-face conversation is the best form of communication.
- Close, daily cooperation between business people and developers.
- Continuous attention to technical excellence and good design.
- Regular adaptation to changing circumstances.
- Even late changes in requirements are welcomed.

Disadvantages of Agile model:

- In the case of some software deliverables, especially the large ones, it is difficult to assess the effort required at the beginning of the softwaredevelopment life cycle.
- There is a lack of emphasis on necessary design and documentation.
- The project can easily get taken off track if the customer representative is not clear what outcome that they want.
- Only senior programmers can take the kind of decisions required during the development process. Hence it has no place for newbie programmers, unless combined with experienced resources.

2.2.2. Project Plan

Task Name	Start	Finish	December				January				February				March			
			1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Plan and Feasibility Study	25/12/2022	09/01/2023				|	|											
Requirements Gathering	10/01/2023	31/01/2023						|	|	|								
Analysis	01/02/2023	07/02/2023									|							
Design	07/02/2023	28/02/2023										|	|	|				
Coding	1/03/2023	31/03/2023													|	| |	|	
Testing	01/04/2023	14/04/2023																

Table 2.2.2

2.2.3. Roles and Responsibilities

Name	Analysis	Frontend	Backend	Testing	Report
Deep Patel	✓	✓	✓	✓	✓
Het Patel	✓	✓	✓	✓	✓
Jenish Patel	✓	✓	✓	✓	✓

Table 2.2.3

3. SYSTEM REQUIREMENTS STUDY

3.1 Study of Current System

The Central Government, under its CRM Scheme, has released more than Rs. 3,062 crores, to the Government of Punjab, NCR State Governments and GNCTD during the five-year period from 2018-19 to 2022-23 towards effective management of stubble in the region. Of the total releases, more than Rs. 1,426 crores have been released to State Government of Punjab.

Regarding availability of machinery for crop residue management procured through the scheme so far, Punjab has about 1.20 Lakh machines: Haryana about 72,700 and U.P. (NCR) about 7,480 machines. During the period, about 38,400 Custom Hiring Centres (CHCs) have been established in Punjab, Haryana, and Uttar Pradesh, of which about 24,200 CHCs are in Punjab and about 6,775 in Haryana.

Based on the Framework advised by the Commission, Action Plans were prepared by the NCR State Governments and Government of Punjab to control stubble burning. For effective monitoring and to ensure integrity of data, the Commission had also formulated a standard protocol for monitoring of paddy residue burning using satellite data with the help of ISRO, IARI and other stakeholders.

The concerted efforts of the Central Government, Government of Punjab, NCR State Governments, and other Stakeholders towards better *in-situ* management of crop residue through the use of CRM machinery, use of PUSA bio-decomposers, facilitating various options for *ex-situ* utilization of paddy straw and extensive IEC activities, educational campaigns, awareness camps and publicity through print, electronic and social media have helped significantly in reducing the fire counts.

3.2 Problems and Weakness of Current System

Current System works fine. It has also some amazing result like it has reduced 31% fire detection (By ISRO) compared to previous years. But as a part of **Digital India initiative,** we have to adopt these types of digital mediums in these efforts. Hence, Effectiveness and Efficiency of these efforts will be going to increase rapidly. Also, we prevent many corruptions because all of transaction will be performed online.

In above efforts sometimes farmers feel laggy because of some issues they have to wait for a long period for harvesting. Also, for auction Government has to conduct auction offline and it will increase the expense compared to these online auctions.

3.3 User Characteristics (Type of users who is dealing with the system)

There are 3 types of users in system require:

FARMERS:

- Register itself and Login
- Request for Service
- View expected revenue

ADMIN**:**

- Register itself and Login
- Analysis of Residue
- Incoming Service request management
- Account Section
- Auction management

Buyer**:**

- Register itself and Login
- Place a bid
- Account Section

3.4 Hardware and Software Requirements (minimum requirements to run your system)

There are no such specific hardware requirements other than basic requirements such as a computer with good internet connectivity and a decent browser which supports React & JavaScript.

Software: -

- Operating System: Windows Operating System 2000 and above and Linux
- Visual Studio Code
- MERN stack
- Mongo database
- QGIS

Mongo database: MongoDB is a Cloud-hosted, NoSQL database that uses a document-model. It can be horizontally scaled while letting you store and synchronize data in real-time among users.

Visual Studio Code: Visual Studio Code is the Integrated Development Environment (IDE) for Web app development.

QGIS: QGIS functions as geographic information system (GIS) software, allowing users to analyse and edit spatial information, in addition to composing and exporting graphical maps. QGIS supports raster, vector and mesh layers.
We haveused it for Supervised Image Classification for raster layers.

3.5 Constraints

3.5.1 Hardware Limitations

- There is only one limitation of this web app, that the device must have a browser.

3.5.2 Reliability Requirements

- The web app does demand much reliability and it is fully assured that the information about the users should be secured and flow is maintained and accessed according to the rights.

3.6 Assumptions and Dependencies

1) Users have sufficient privileges to access the internet.
2) Browser on Device is running smoothly.
3) Database updates give expected and accurate results.

4. SYSTEM ANALYSIS

4.1 Requirements of New System (SRS)

4.1.1 User Requirement

1. Farmers
2. Admin/Government
3. Buyer's

1. Farmers

- Manage Profile: Allows farmers to view and update his/her details.
- Register new crop intake: Allow farmers to update current crop intake.
- Request for service: Allows farmers to apply for harvesting and assembling related services.

2. Admin/Government:

- View Farmers profile: Allows admin to view farmers crop intake.
- Manages residue stock: Admin has to keep eye on stock, so they got a acknowledgement for auction.
- View service requests: Allows admin to view incoming service requests providedby farmers and manage it.
- Declare Auction: Allows admin to declare auction and post it on website.
- Auction Management: Allows admin to manage auction in fair way and at end of deadline they must declare results.
- Manages Account section: Allows admin to manage all the payment related stuffs.

3. Buyers

- Registration: Allows Buyers to register them for auction.
- Placing a bid: Allow companies to place a bid during the auction period.
- Account section: Buyers must manage their own account and payment related stuffs with Government.

4.1.2 System Requirements

1. Functional requirements

1) Manage user's identity.

1.1- Differentiating User's

- **Description: -**To differentiate if the User is a farmer or a buyer, We have to give an option to select from a farmer And buyer.
- **Output: -**Different forms for farmers and buyers will appear.

1.2- Registration for farmers

- **Input:** -Name, Mobile number, Bank Account details, land Location, Land size, Email, password
- **Output: -** Create a new user as a farmer.

1.3- Registration for Buyers

- **Input: -**Name, Mobile number, company or individual, Email, password
- **Output: -** Create a new user as a Buyer.

1.4- Login for existing user

- **Input: -**Mobile number / Email & password
- **Output: -** If mobile number and email matches with the existing user data then check for a password if match with existing user's password allow login, else show error massage.

2. Creating a request for harvesting of crops: - This request for harvesting Should be made earlier and specify the duration in which the harvesting Must be done according to the health of crops.

- **Input: -**Types of crops, Amount of land specific crops are planted, Time duration for harvesting, numbers of labours and machines needed for harvesting.

- **Output: -** According to the availability of a machines and labours Estimated date for harvesting will be shown after request is sent, and according to the time slot labours and government bodies will be informed about a request.

3. Purchasing of residue and grains:

3.1- Purchase Residue

- **Input: -**Type of Residue they want to buy, Amount of residue they want, Address where they want residue at
- **Output: -** This purchase should be placed, and residue data should be updated Estimated date on which they Residue will be Delivered.

3.2- Purchase Grain: -

Here option for buying will be according to the Farmers as they will select the Buyer to sell Their grains. So, they will have to give a price at which they want to buy the grains beforehand.

- **Input: -**Type of Grains they want to buy, Number of grains they want, Address where they want Grains at

- **Output: -** Request to buy the grain should be updated and buyer Should be listed in the buyers list with the type to grain they want and price.

4. Feedback for the harvesting service-

Farmers should have an option of giving feedback of the service they will get from the government bodies.

- **Input: -** Answers to feedback questions

- **Output: -** Feedback of farmer should be stored and according to The rating the government employee should be remarked.

5. Raising issue/complaint-

Users should have a option for raising of issue against service they get.

- **Input: -** Detailed Description of an issue/Complaint.

- **Output: -** Complaint of farmer should be stored and according To the issue steps should be taken to resolve the issue

2. Non-Functional Requirements

2.1 Usability:

The system must be easy to use by the admin such that they do not need to read an expensivenumber of manuals.

The options of the system must be easily navigable by the admin with buttons that are easy to understand.

2.2 Performance:

- The responsiveness of the website shall be high, and the website shall behave as per the user action.
- The user shall be acknowledged in the form of visual changes or feedback on the site to enhance the interaction
- The response time and throughput time on the site shall be minimal
- Consistency on the website shall be maintained across all the web pages
- The layout of the site shall be kept simple and must be self-explanatory

2.3 Safety Requirements:

2.3.1 If there is extensive damage to a wide portion of the database due to catastrophic failure, such as a disk crash, the recovery method restores apast copy of the database that was backed up to archival storage (typically tape) and reconstructs a more current state by reapplying or redoing the operations of committed transactions from the backed-up log, up to the time of failure.

2.4 Security Requirements:

- The website shall offer secure login option to the users to avoid unauthorized access to the system and the information
- Advanced access control shall be included in the site
- Advanced encryption algorithms must be integrated in the site to avoid misuse of the data sets.
- Technical controls, such as anti-malware, anti-denial, and intrusion detection tools shall be integrated with the site.

5. System Design

5.1. Use Case Diagram

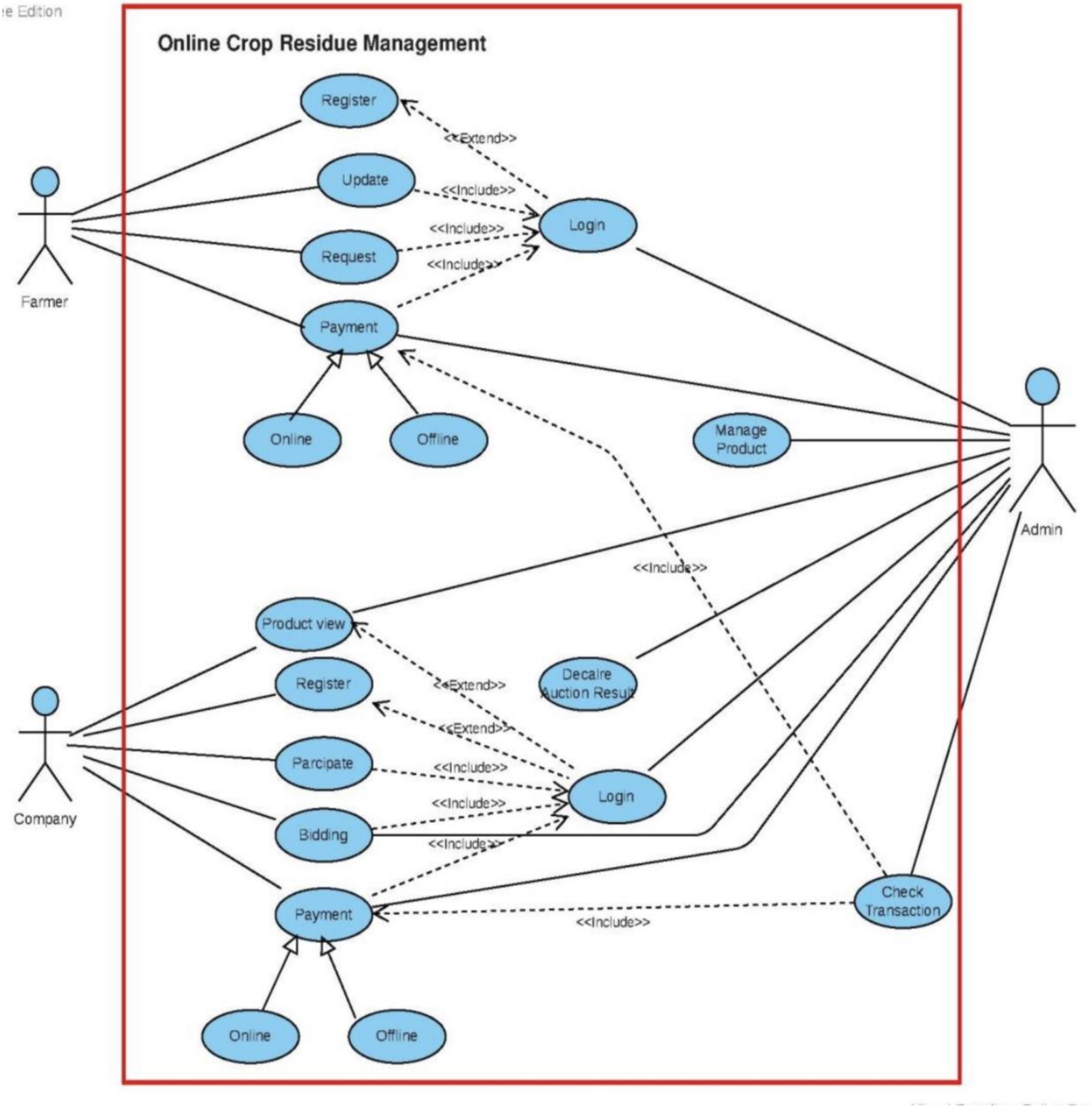


Fig 5.1

5.2. Class Diagram

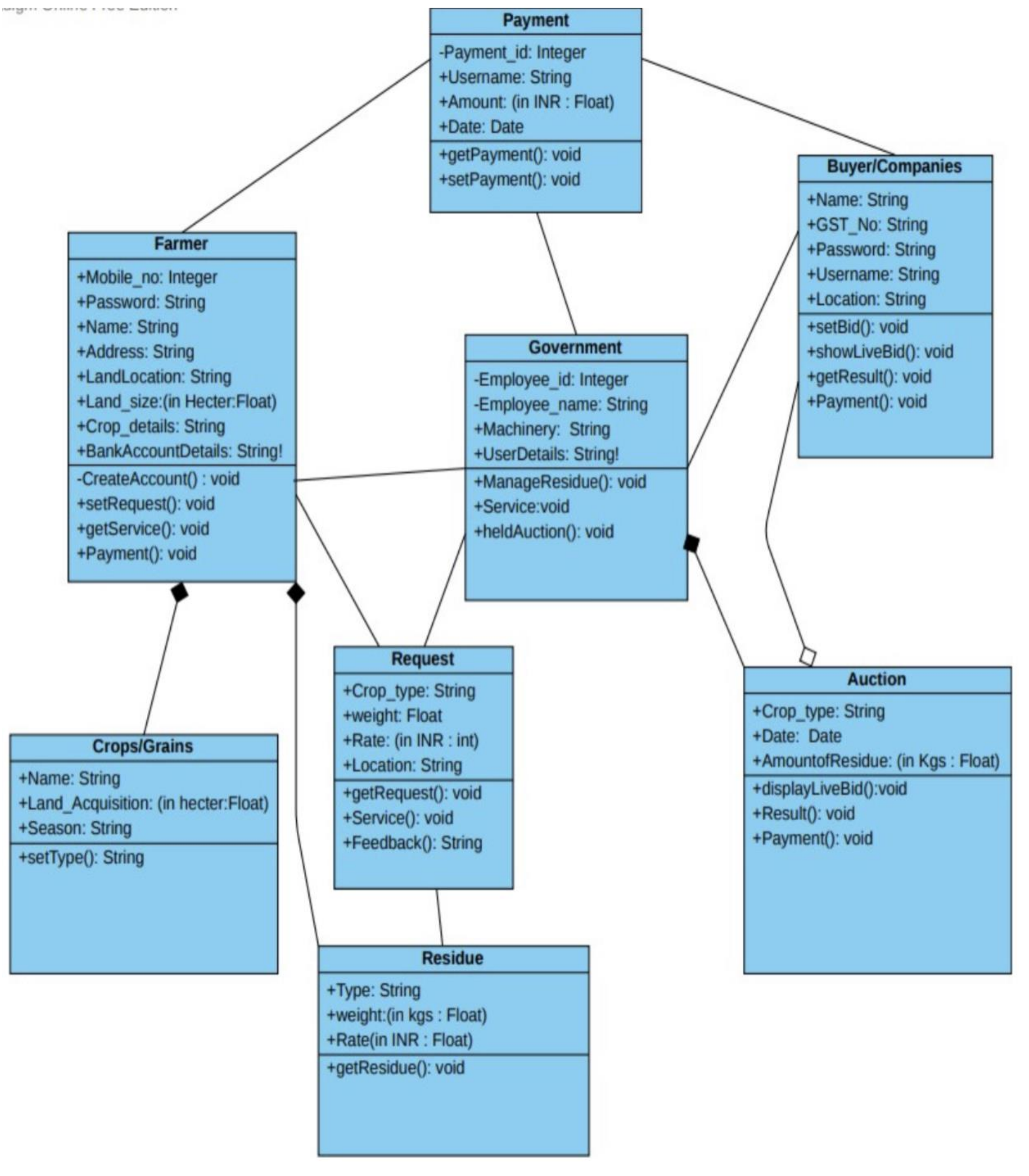
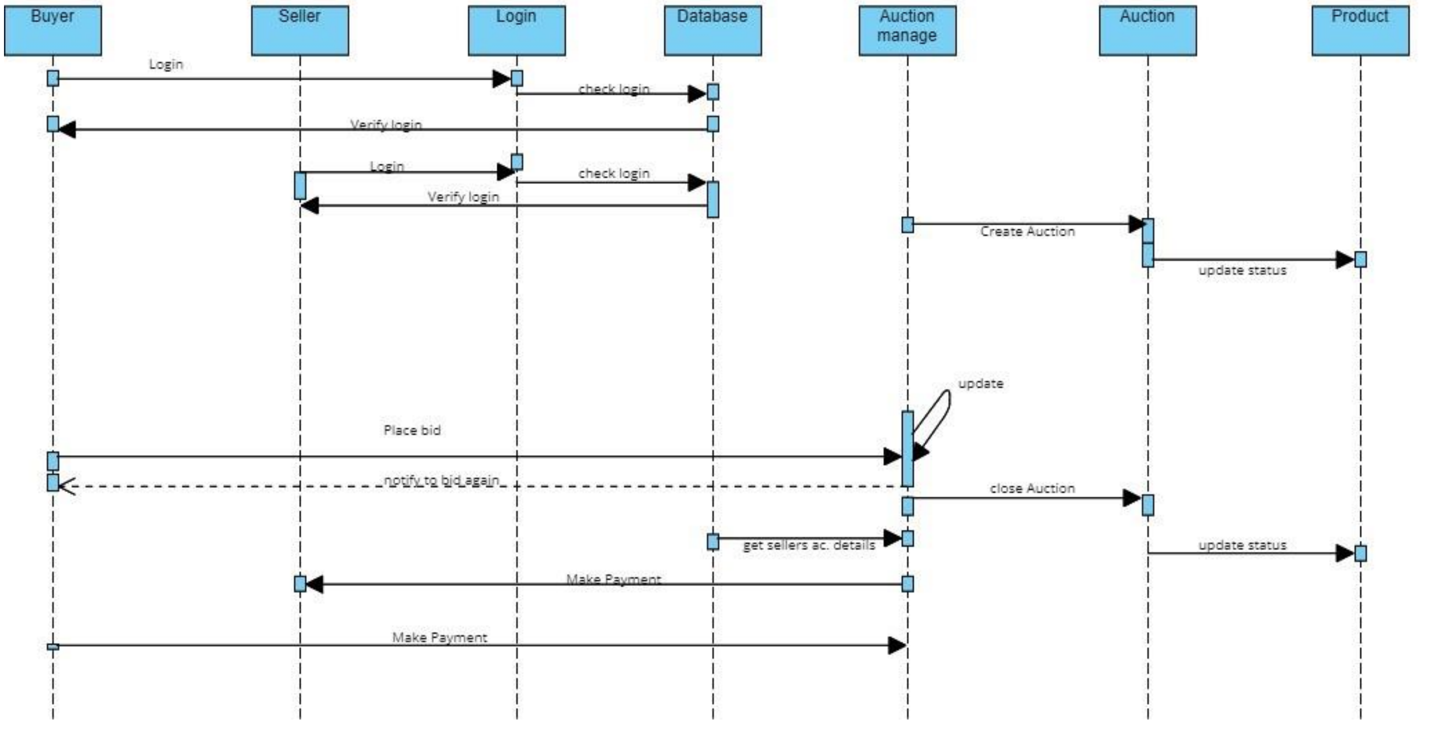


Fig 5.2

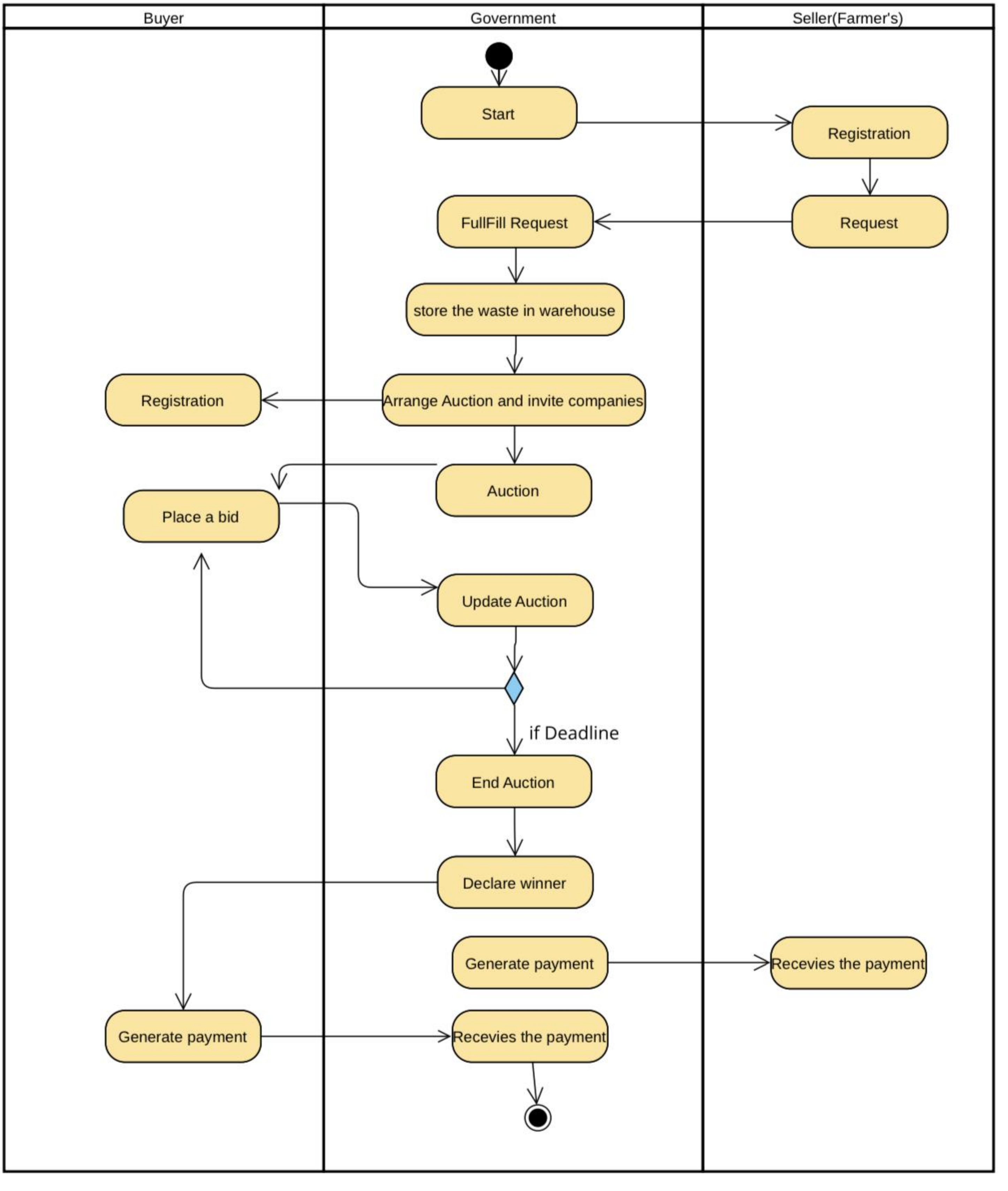
5.3. Sequence Diagram



(Sequence Diagram**)**

Fig 5.3(a)

5.4. Activity Diagram



(Activity Diagram)Fig 5.4

6. IMPLEMENTATION PLANNING

6.1 Implementation Environment (Single vs Multi User, GUI vs non-GUI)

For implementation we have used:
Visual Studio Code
Our project is built using Visual Studio Code. Seeing that it is an web application we saw fit that Visual studio code provided us with all the required basis for successful implementation of our web app. Also to for storing our data we have used Mongo database which enables our web app to run at all times.

6.2 Program/Modules Specification

The following Modules are implemented:

- Farmer
  - Manage Profile
  - Register for current Crop intake.
  - Request for Service (Harvesting, Assembling)
- Government
  - Manages farmers requests and deals with end workers.
  - Manages residue stock.
  - Declare Auction
  - Manages auction and at the end of deadline declares results.
  - Manages account section.
- Buyers
  - Register to take part in the auction.
  - Placing a bid (Harvesting, Assembling)
  - Manages self-account section.

6.3 Coding Standards

To make the system coding easy, easy to remember and reduce the chances of errors, some techniques are used at the time of coding of the application which is called coding standard. The coding standard which we adopted during the coding is explained as follows:

- Each nested block should be properly indented and spaced.
- The code should be properly commented on so that we can understand it easily. Comments regarding the statements increase the understandability of the code.
- Better to avoid the use of digits in variable names.
- The names of the functions should be written in camel case starting withsmall letters.
- The name of the function must describe the reason for using the function clearly and briefly.

7. TESTING

7.1 Testing Plan

The testing technique that is going to be used in the project is White box testing. In White box testing the Tester has knowledge about the internal structure of the code or the program of the software.

7.2 Testing Strategy

The development process repeats this testing subprocess several times for the following phases.

a) Unit Testing.

b) Integration Testing

Unit Testing tests a unit of code (module or program) after coding of that unit is completed.

Integration Testing tests whether the various programs that make up a system, interface with each other as desired, fit together and whether the interfaces between the programs are correct.

Testing is carried out in such a hierarchical manner to ensure that each component is correct, and the assembly/combination of components is correct. Merely testing a whole system at the end would most likely throw up errors in components that would be very costly to trace and fix.

7.3 Testing Methods

Black Box and White Box Testing:

In black box testing a software item is viewed as a black box, without knowledge of its internal structure or behavior. Possible input conditions, based on the specifications (and possible sequences of input conditions), are presented as test cases.

In white box testing knowledge of internal structure and logic is exploited. Test cases are presented such that possible paths of control flow through the software item are traced. Hence more defects than black-box testing is likely to be found.

Out of the 2 methods for testing, black box testing and white box testing, we would be using the white box testing as we are well aware of the internal functionalities of our application unlike in the black box testing, where we require a 3rd party to test our cases and the internal details are hidden from him.

7.4 Test Cases

1.Farmer

Entity	Test Case	Expected Output	Actual Output	Result
Registration of Farmer	Validation	Successfully Registered	Successfully Registered	Pass
Login of Farmer	Validation	Login Successful	Login Successful	Pass
Profile	View Profile	View Profile Details Successful	View Profile Details Successful	Pass
Create Service	Add Crops details and specific details Required.	Service Requested successfully	Service Requested successfully	Pass
View Research Page	Click on Learn More Button	Research Page Displayed successfully	Research Page Displayed successfully	Pass
Enter Data in Form of Research Page	Enter Size of Land in Input field	Data Displayed successfully	Data Displayed successfully	Pass

Table 7.4.1 Farmer Test Case

2. Buyer

Entity	Test Case	Expected Output	Actual Output	Result
Registration for Buyer	Validation	Successfully Registered	Successfully Registered	Pass
Login for Buyer	Validation	Login Successful	Login Successful	Pass
Enter in Particular Auction Room	Validation	Entered In Auction Room Successfully	Entered In Auction Room Successfully	Pass
Submit a Bid in Auction	Validation	Bid Submitted Successfully	Login Successful	Pass
Show Auction Result	Validation	Show Auction Result Successfully	Show Auction Result Successfully	Pass

Table 7.4.2 Buyer Test Case

3.Admin

Entity	Test Case	Expected Output	Actual Output	Result
Registration of Admin	Validation	Successfully Registered	Successfully Registered	Pass
Login of Admin	Validation	Login Successful	Login Successful	Pass
Create Auction Room	Enter required Data	Auction Room Created successfully	Auction Room Created successfully	Pass
View Room Details	Click on Any service from Pending Service List in Admin HomePage	View Room Details successfully	View Service Details successfully	Pass
View Service Details	Click on Any service from Pending Service List in AdminHomePage	View Service Details successfully	View Service Details successfully	Pass
Fulfill Request Form	Enter Required Data	Request Fulfilled successfully	Request Fulfilled successfully	Pass

Table 7.4.3 Admin Test Case

8. USER MANUAL

User Manuals are manuals that enable the user of a system or application to understand the working of the system and help them to use them efficiently. It is usually written by a technical writer, although user guides are written by programmers, productor project managers, or other technical staff, particularly in smaller companies.

Follow below mentioned steps in order to work with the app.

8.1 WEB-DEVLOPMENT

Home page

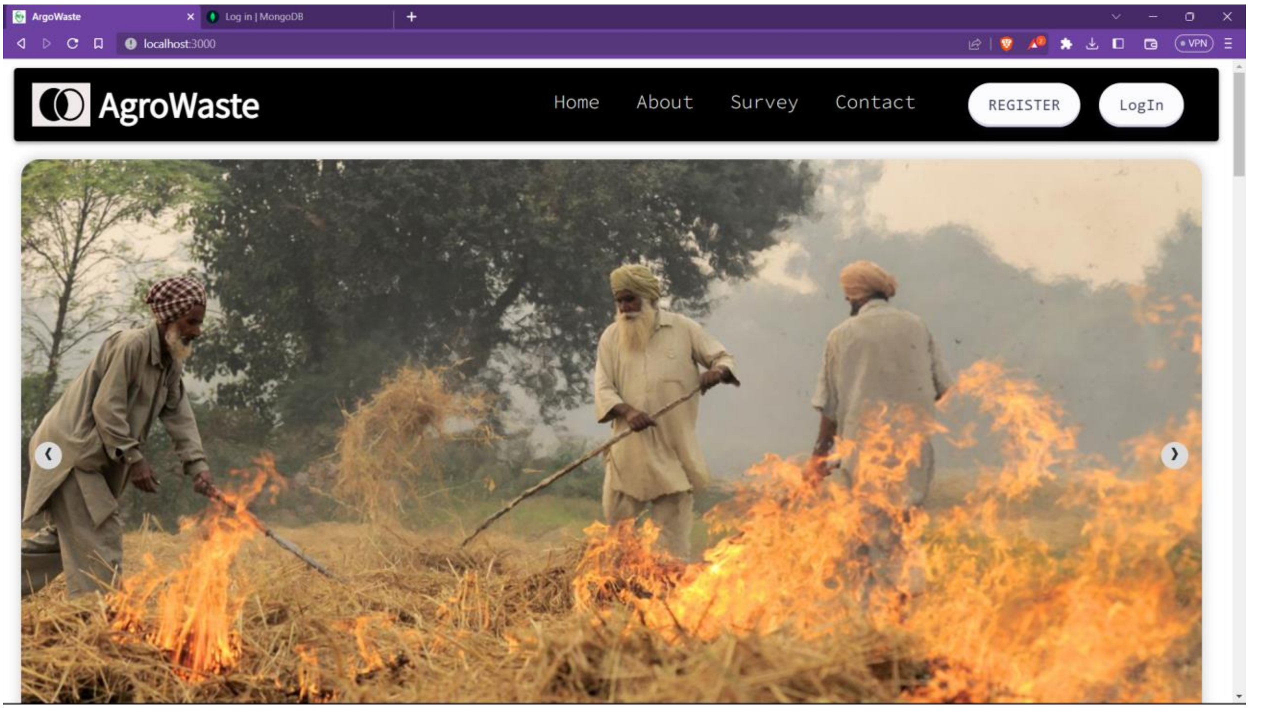


fig 8.1

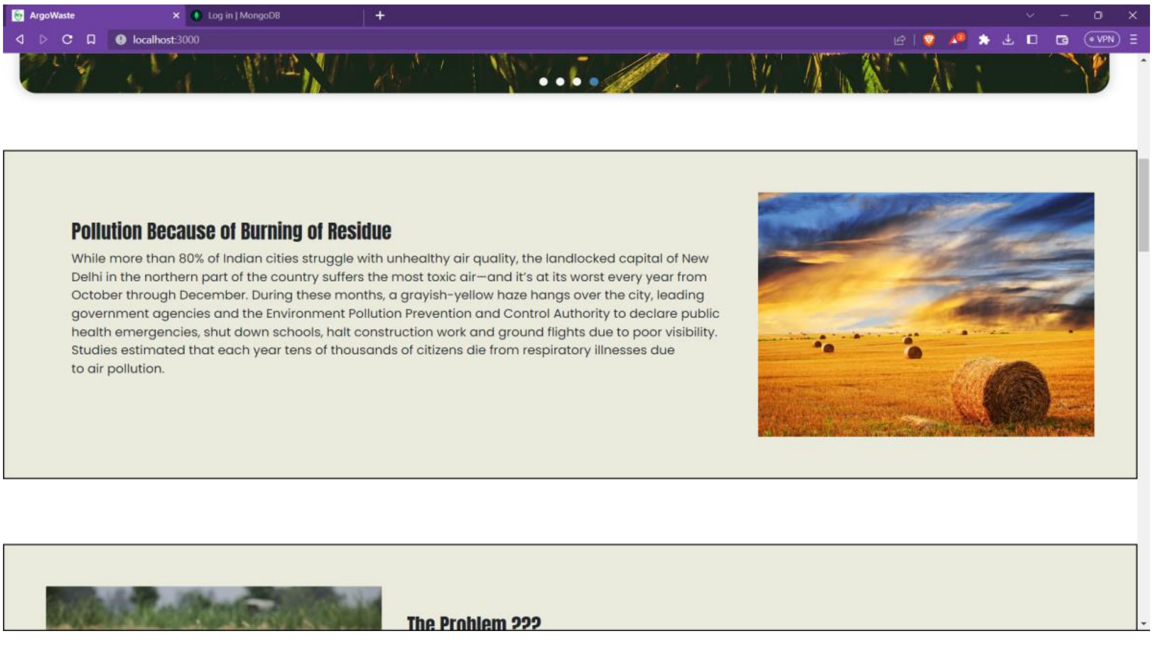


fig 8.2

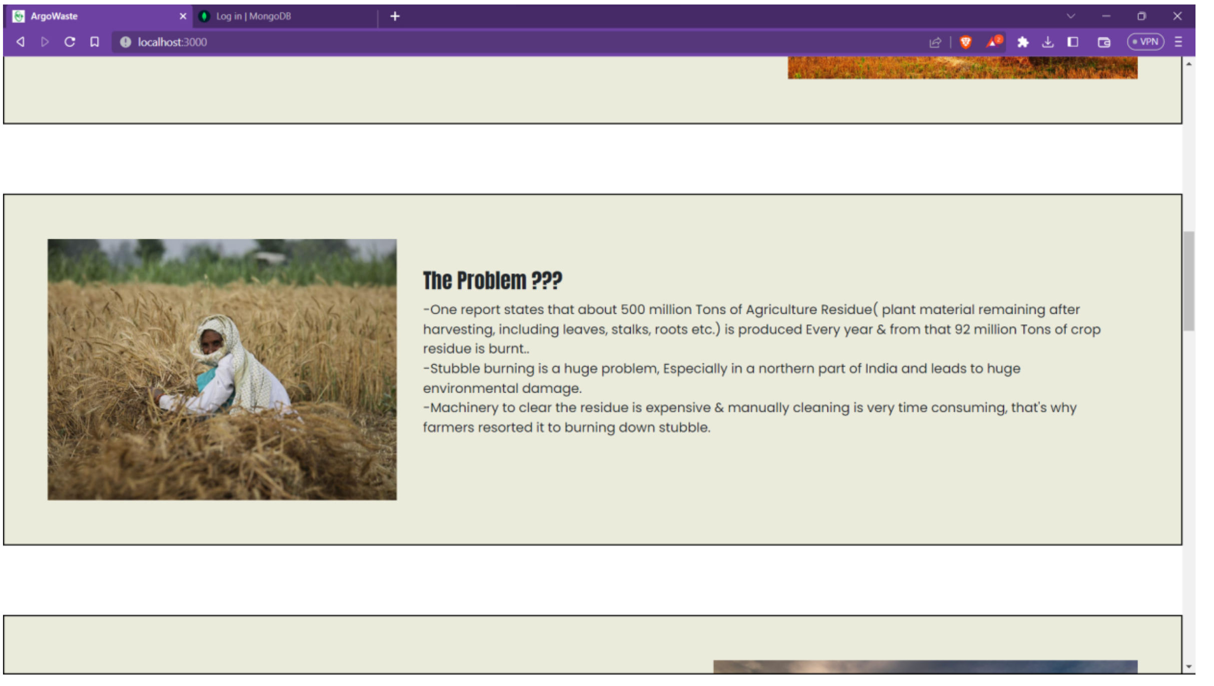

ArgoWaste
Log in | MongoDB
localhost:3000
The Problem ???
-One report states that about 500 million Tons of Agriculture Residue(plant material remaining after harvesting, including leaves, stalks, roots etc.) is produced Every year & from that 92 million Tons of crop residue is burnt..
-Stubble burning is a huge problem, Especially in a northern part of India and leads to huge environmental damage.
-Machinery to clear the residue is expensive & manually cleaning is very time consuming, that's why farmers resorted it to burning down stubble.

fig 8.3

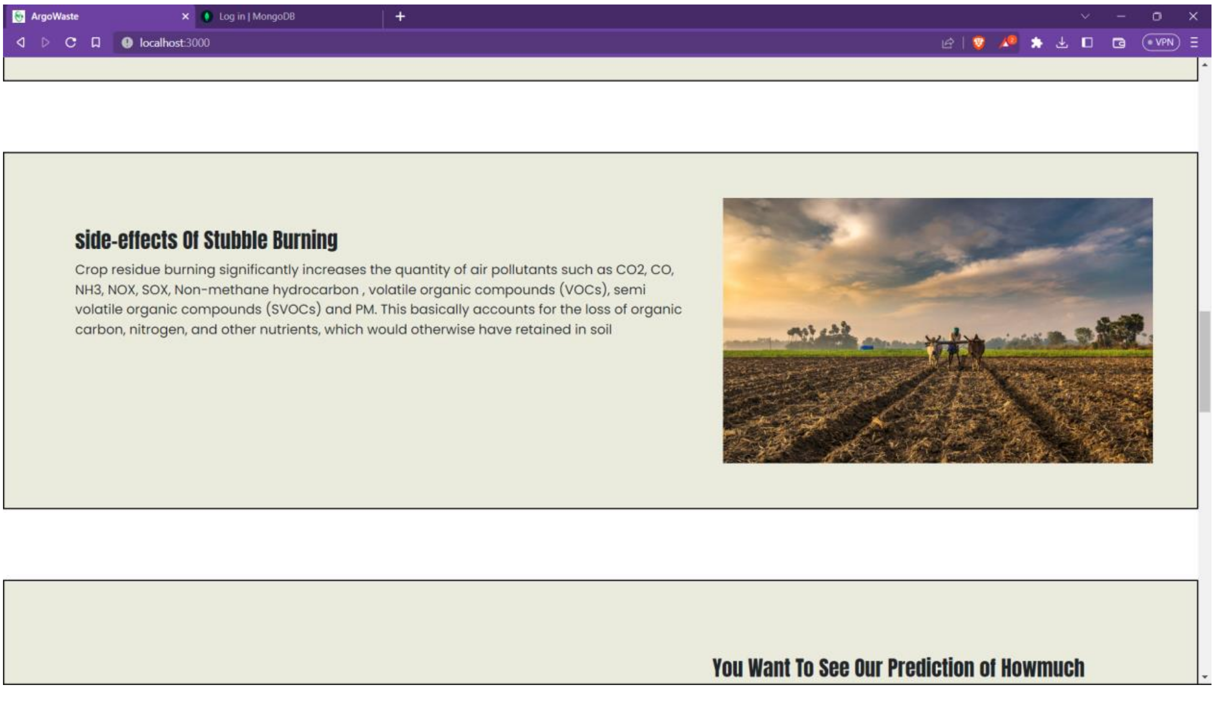

ArgoWaste
Log in | MongoDB
localhost:3000
side-effects Of Stubble Burning
Crop residue burning significantly increases the quantity of air pollutants such as CO2, CO, NH3, NOX, SOX, Non-methane hydrocarbon , volatile organic compounds (VOCs), semi volatile organic compounds (SVOCs) and PM. This basically accounts for the loss of organic carbon, nitrogen, and other nutrients, which would otherwise have retained in soil
You Want To See Our Prediction of Howmuch

fig 8.4

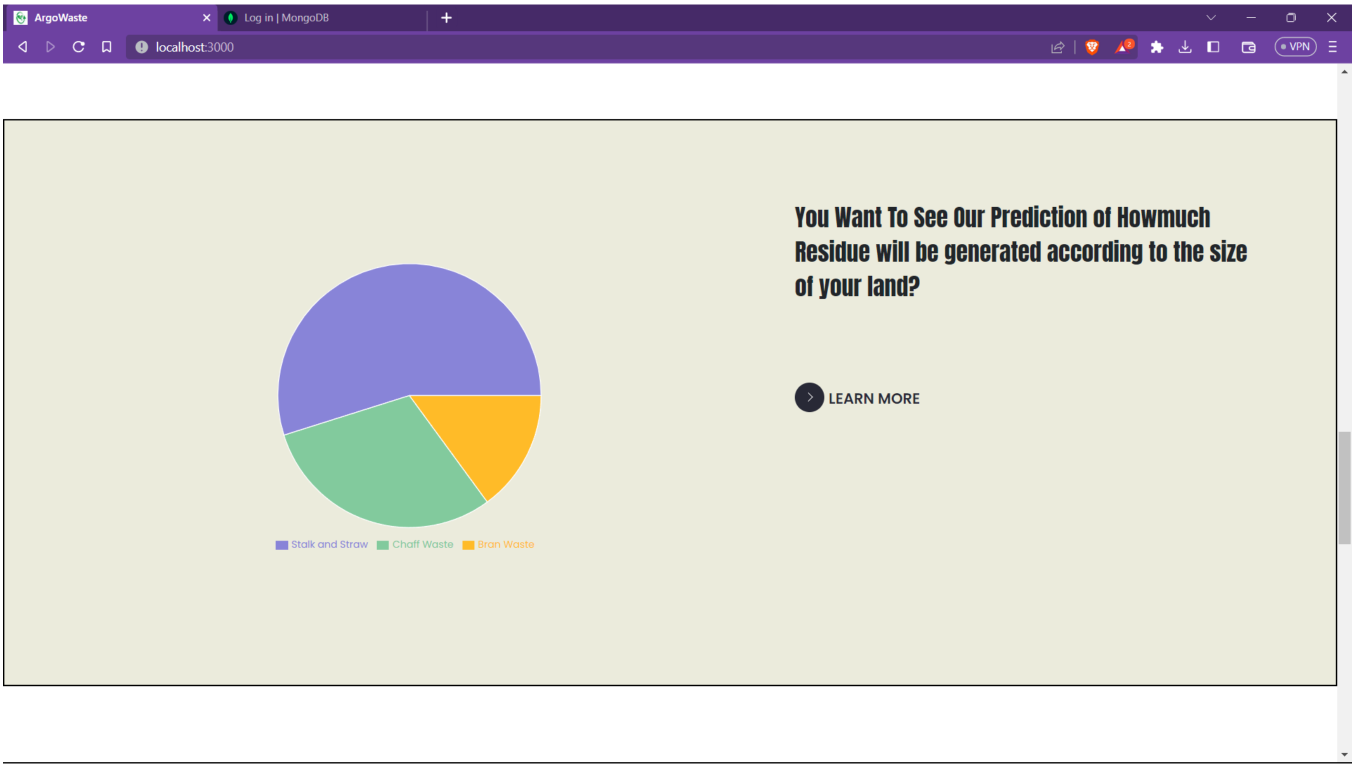

ArgoWaste
Log in | MongoDB
localhost:3000
VPN
You Want To See Our Prediction of Howmuch Residue will be generated according to the size of your land?
LEARN MORE
Stalk and Straw
Chaff Waste
Bran Waste

fig8.5

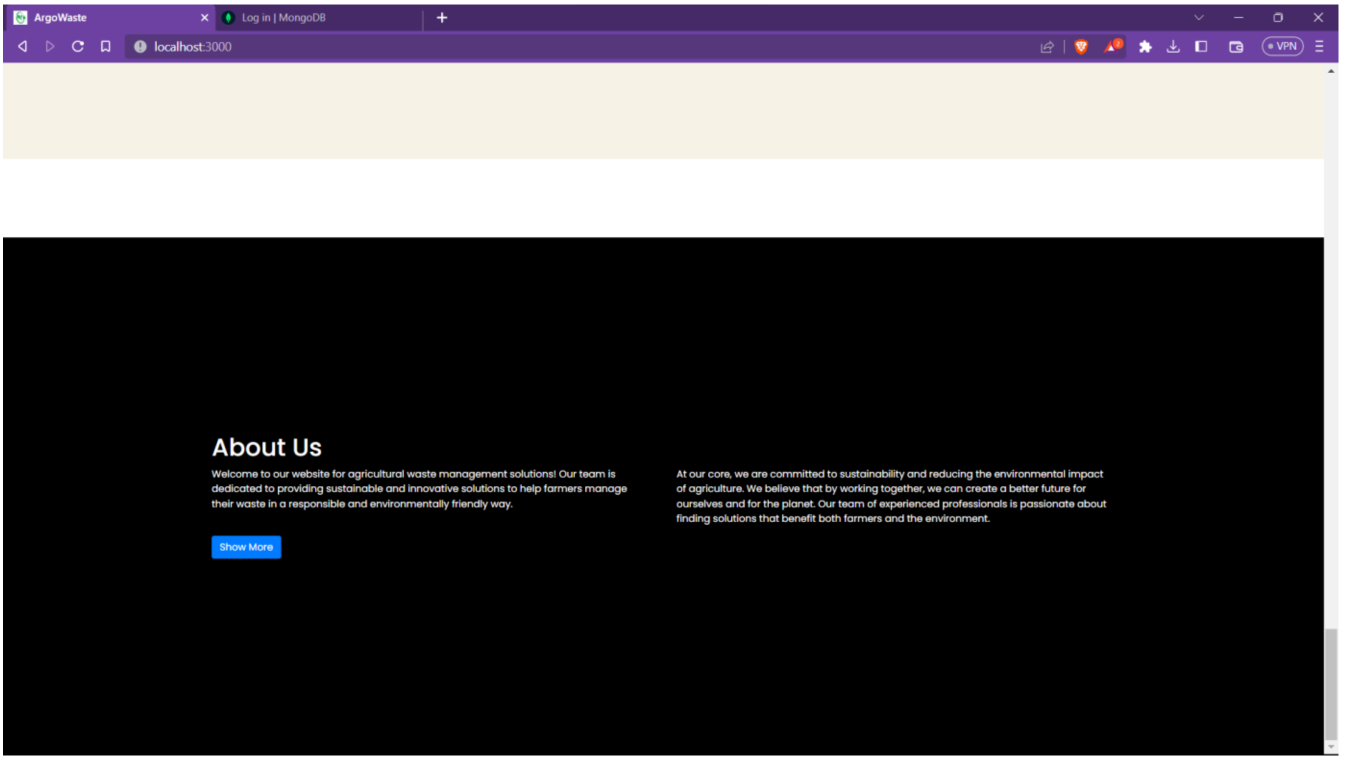

ArgoWaste
Log in | MongoDB
localhost:3000
VPN
About Us
Welcome to our website for agricultural waste management solutions! Our team is dedicated to providing sustainable and innovative solutions to help farmers manage their waste in a responsible and environmentally friendly way.
Show More
At our core, we are committed to sustainability and reducing the environmental impact of agriculture. We believe that by working together, we can create a better future for ourselves and for the planet. Our team of experienced professionals is passionate about finding solutions that benefit both farmers and the environment.

fig 8.6

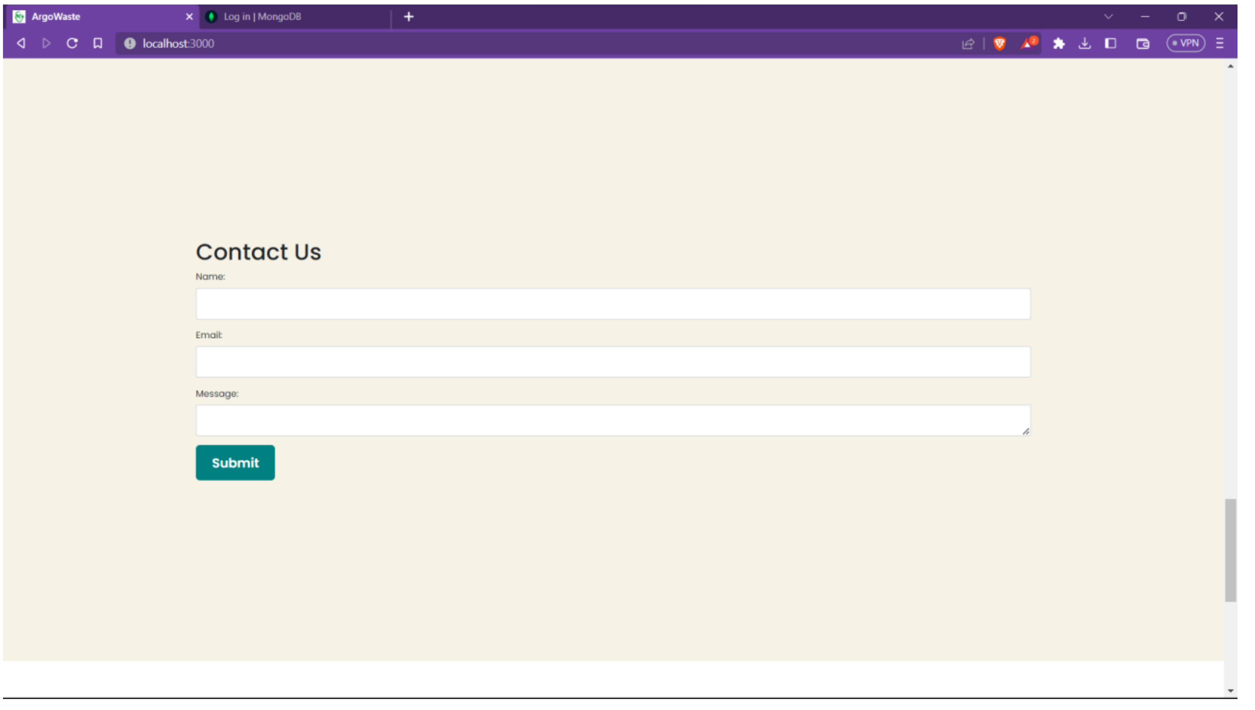


fig 8.7

Registration panel

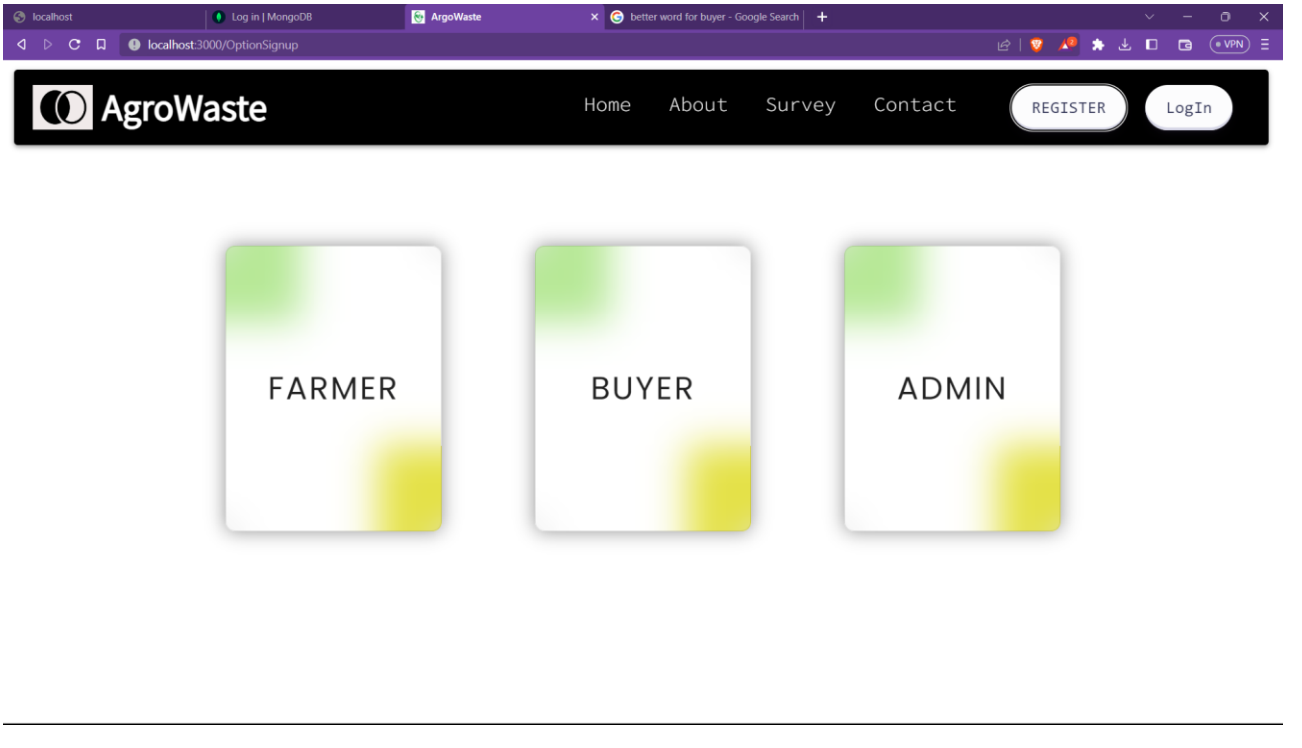
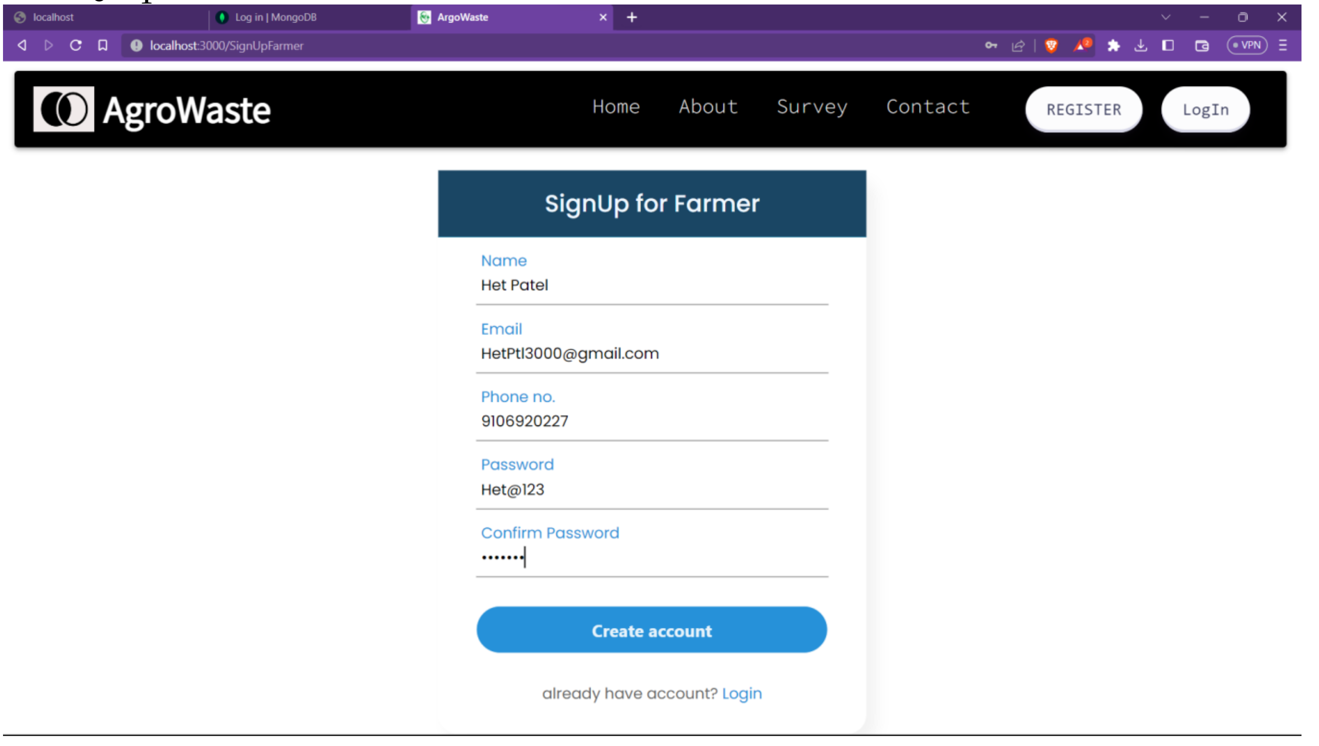


fig 8.8

Farmer panel

1. Registration:

- Farmer will be registered by himself/herself only.
- T



g

Fig 8.9 Registration by farmer

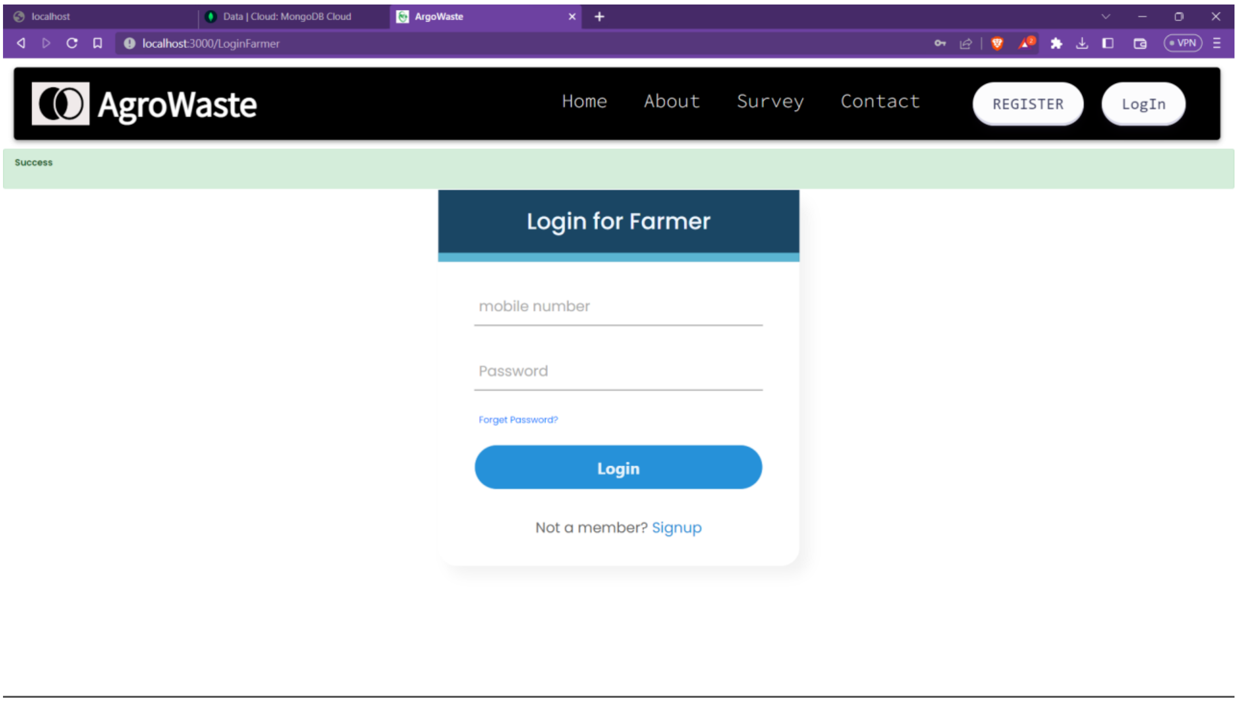


Fig 8.10 Redirected to Login page.

2. Login:

- Farmers would just have to be registered once else you will be simply logged into the website

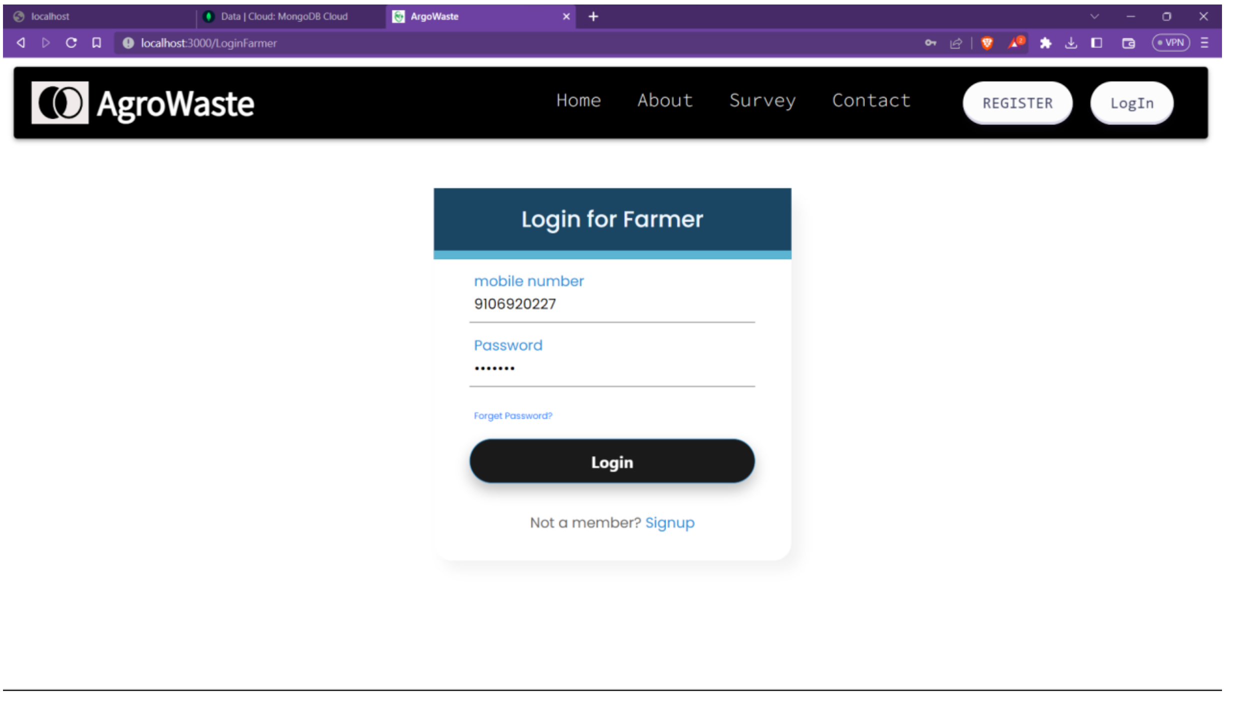


fig 8.11 Farmer Login

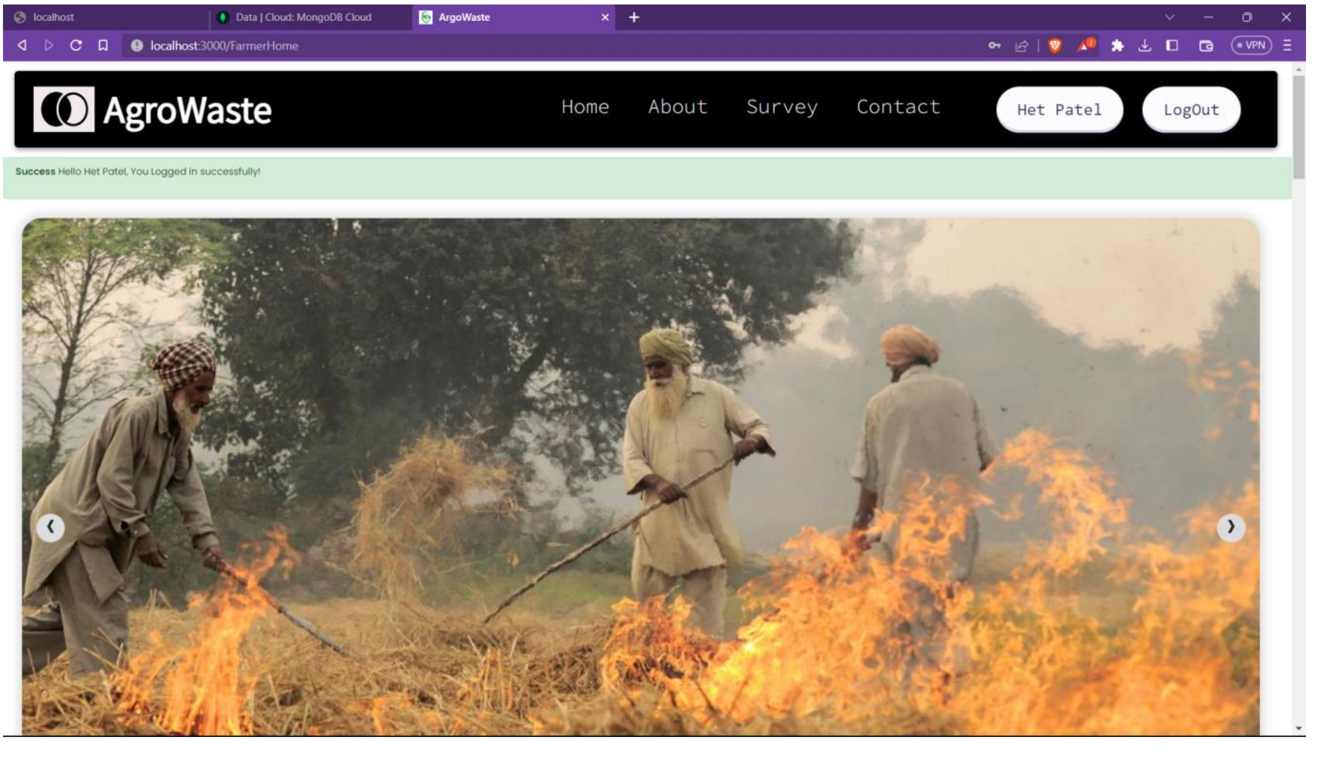


fig 8.12 Redirect to home page

3. Service Request:

- Landing page of Farmer panel. After successful login farmer will request for service from this page
- On the service page farmers has to fill the details related to their crop and also give expected harvesting duration so, that they will get service in that period only
- Farmers also has two choices, Either they want to sell residue only or residue and crop both.

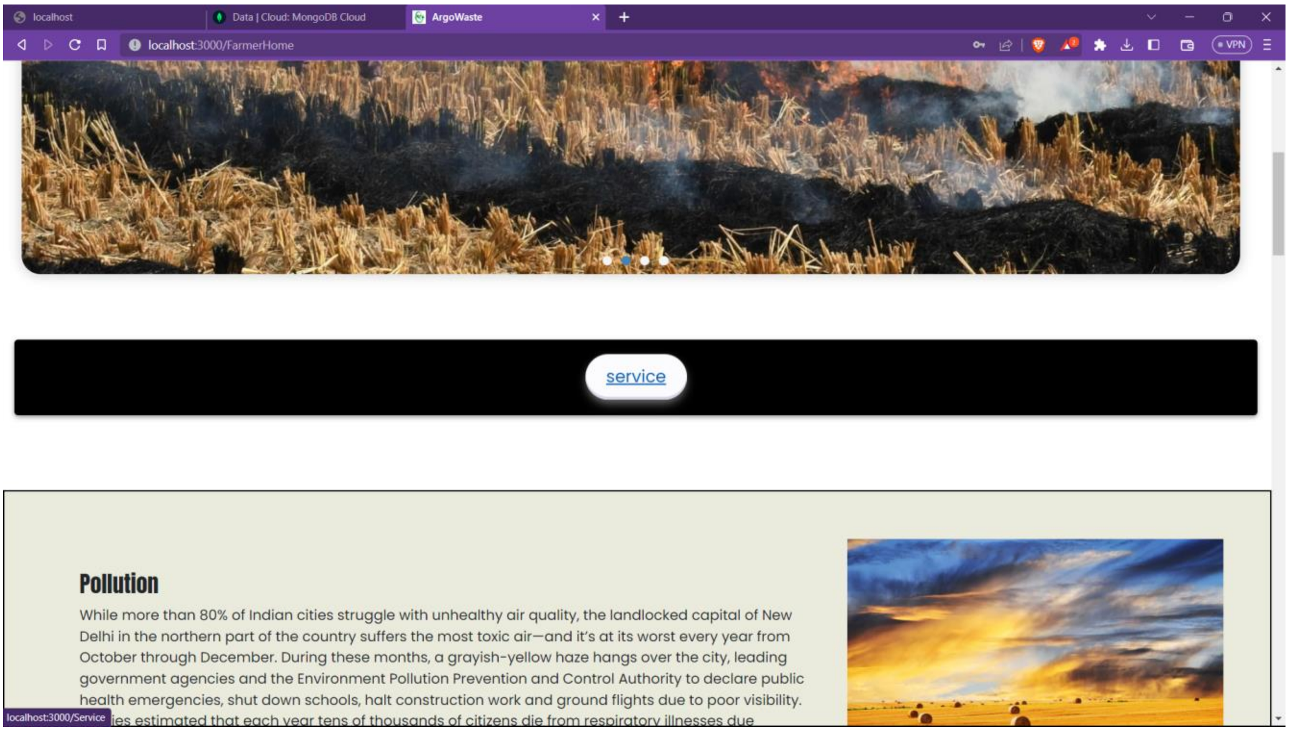


Fig 8.13 Service

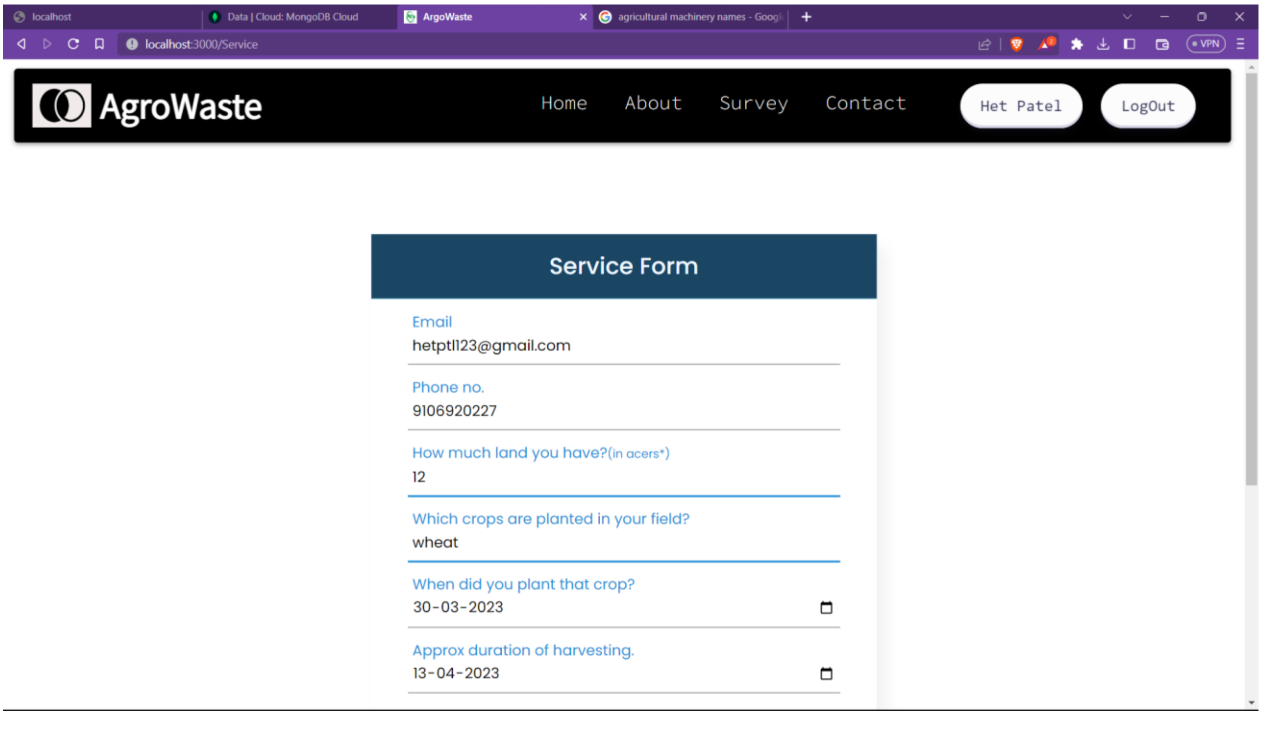


Fig 8.14 Service Page

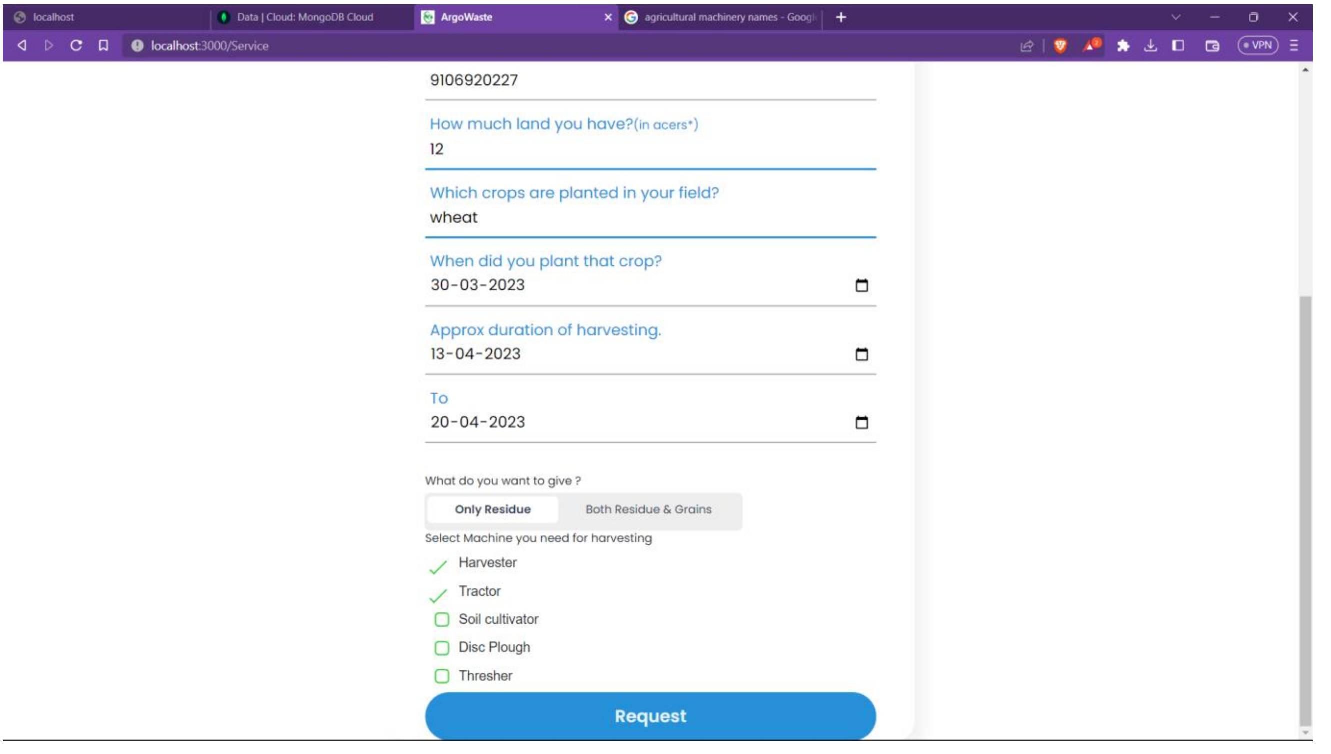


Fig 8.15 Service Page

- Farmer's Profile

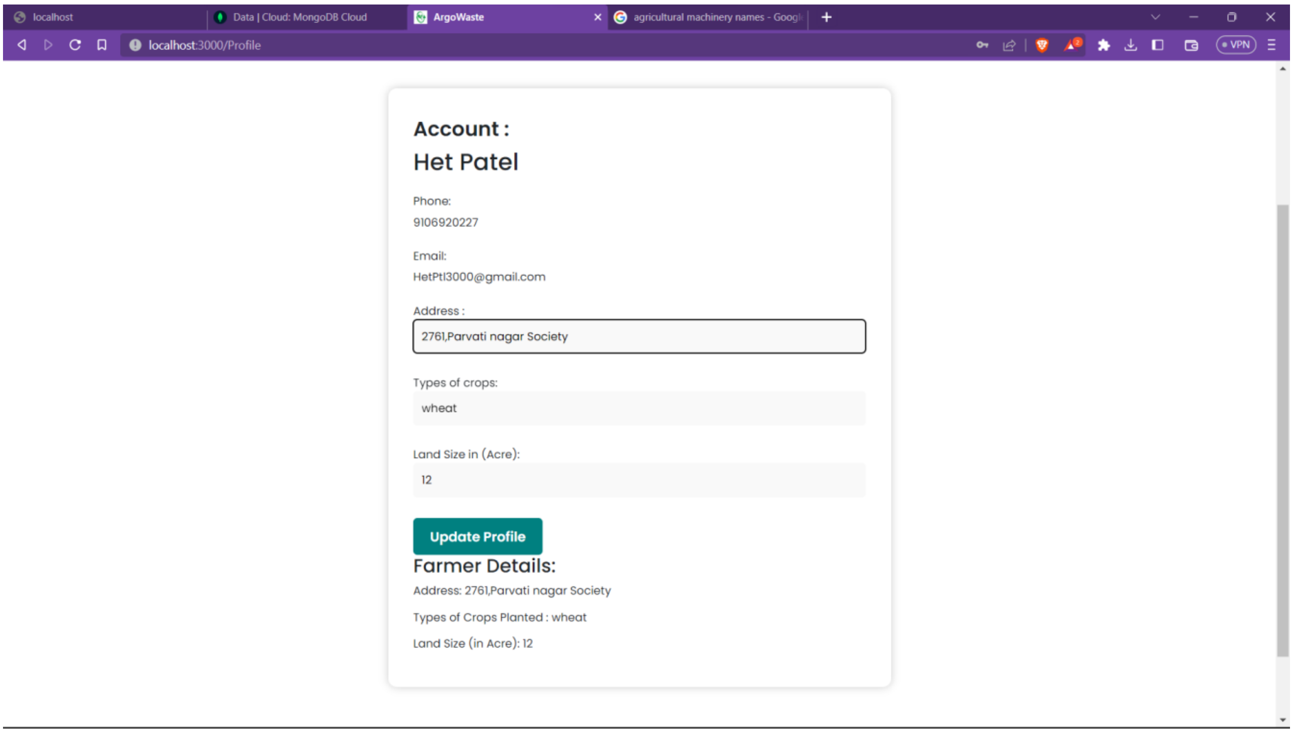


fig 8.16 Farmer profile

Buyer’s Panel

1. Registration:

- Buyers/Companies will be registered by himself/herself only.
- Then they would be redirected to Login page.

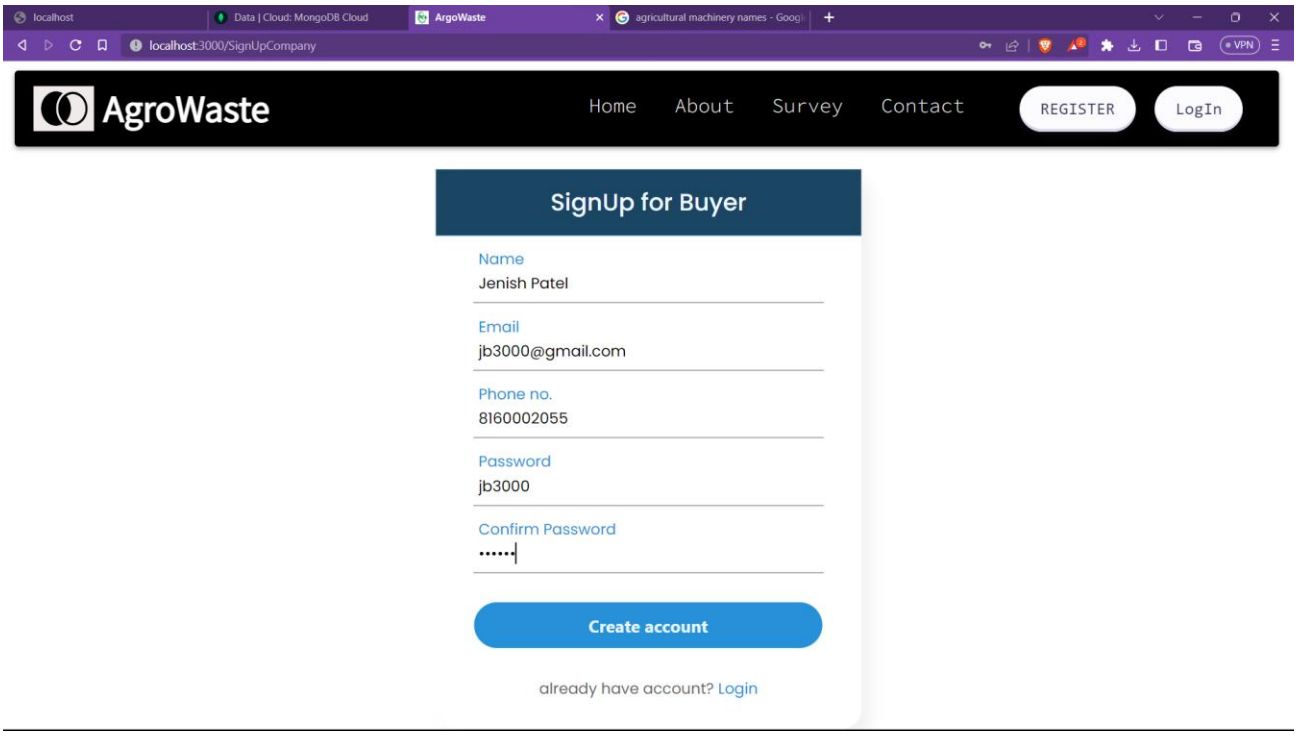


Fig 8.17 Buyer Signup

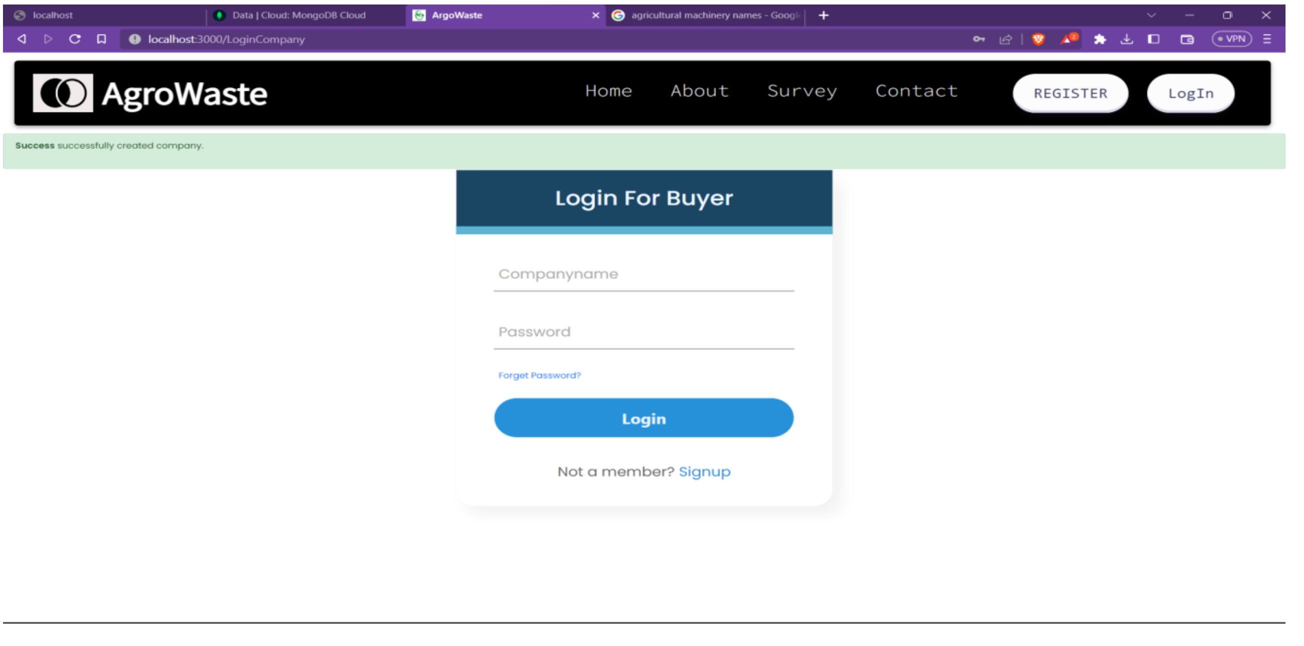


Fig 8.18 Buyer redirect to login page

- Login For Buyer

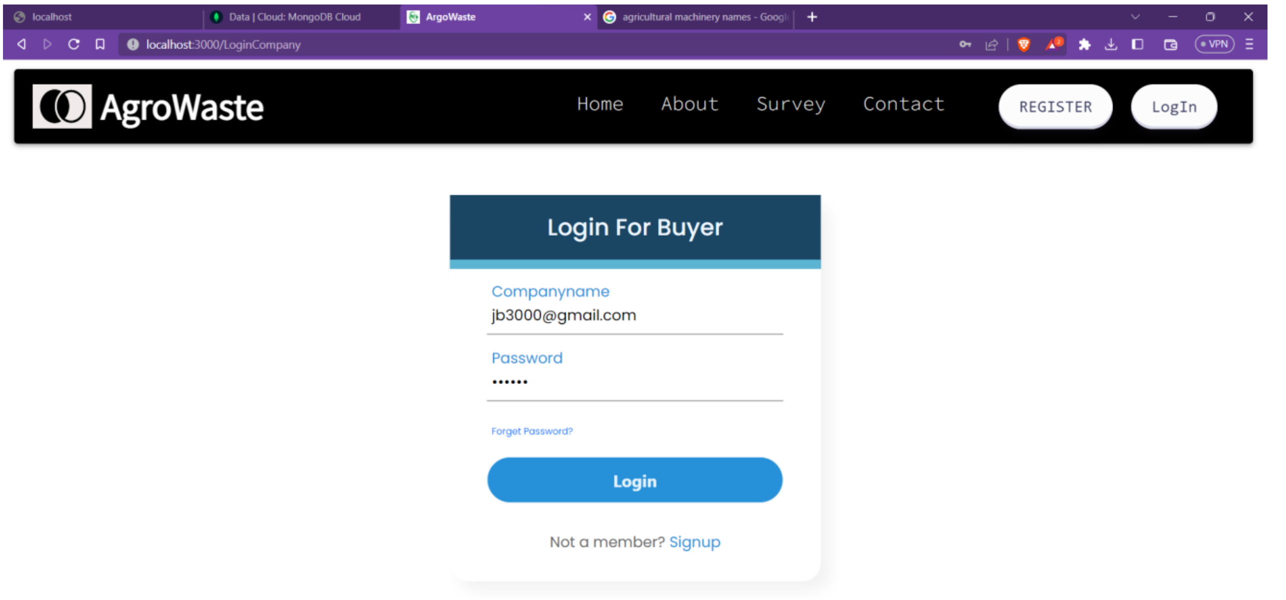


Fig 8.19 Buyer login

-

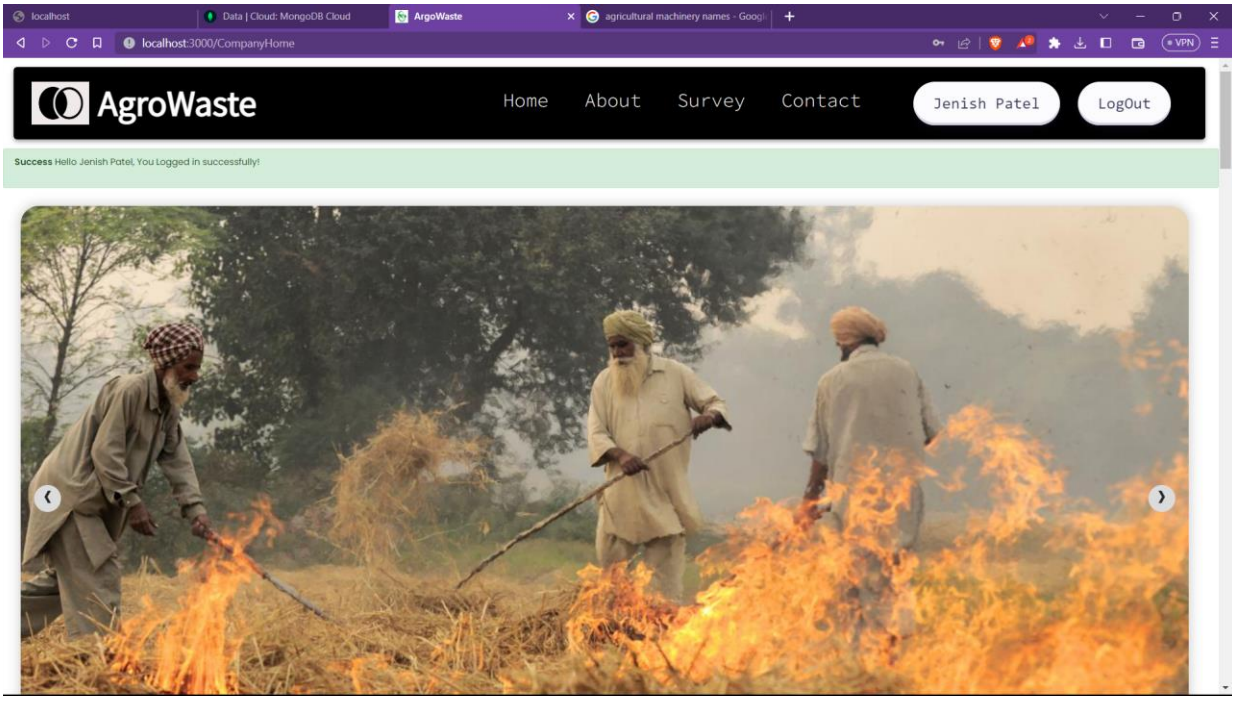


Fig 8.20 Buyer redirect to home page

2. Auction:

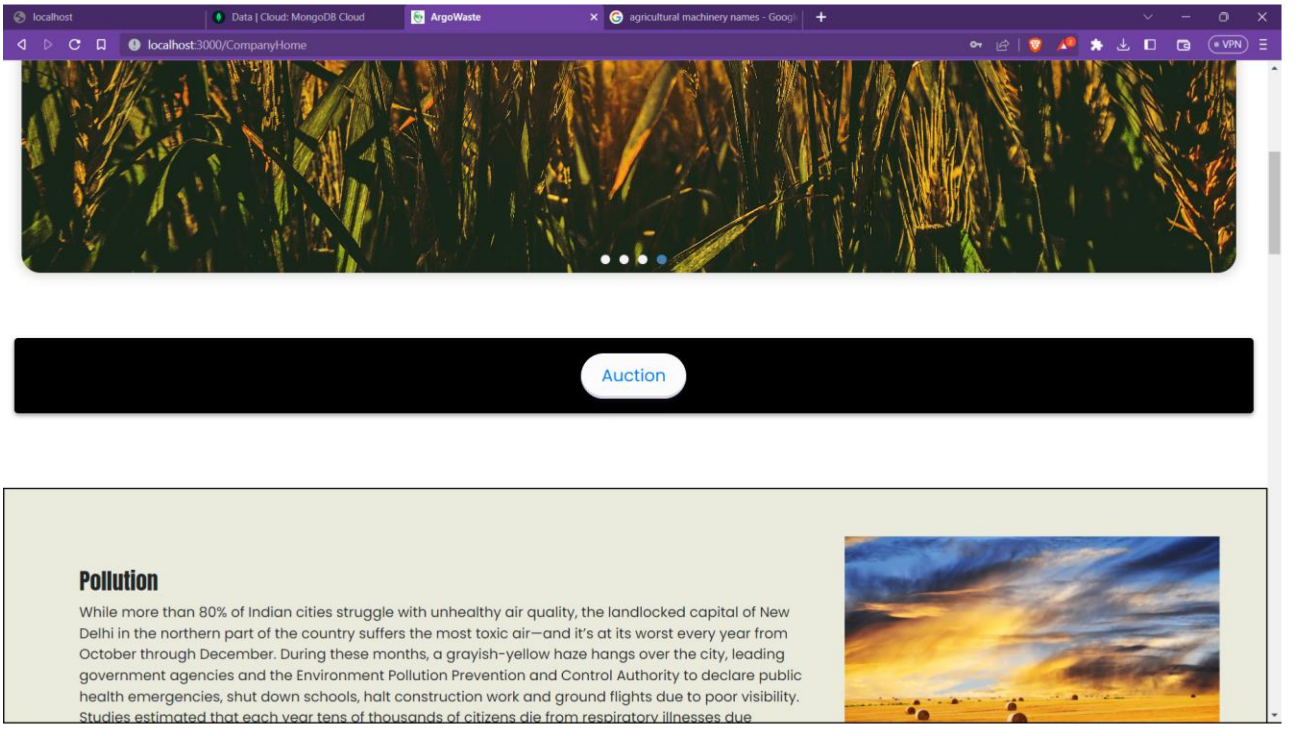


Fig 8.21 Buyer's Auction option

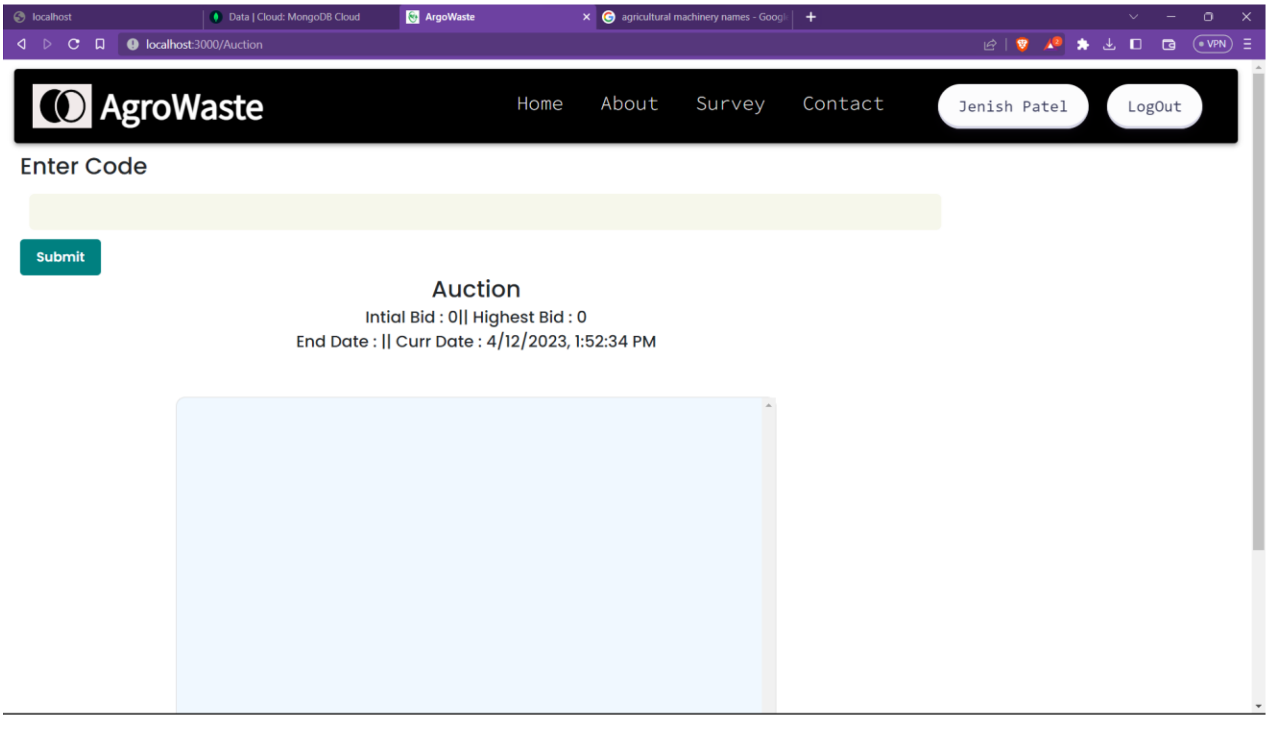


Fig 8.22 Auction Page

- Unique code for auction room entry which will be provided by Government .
-

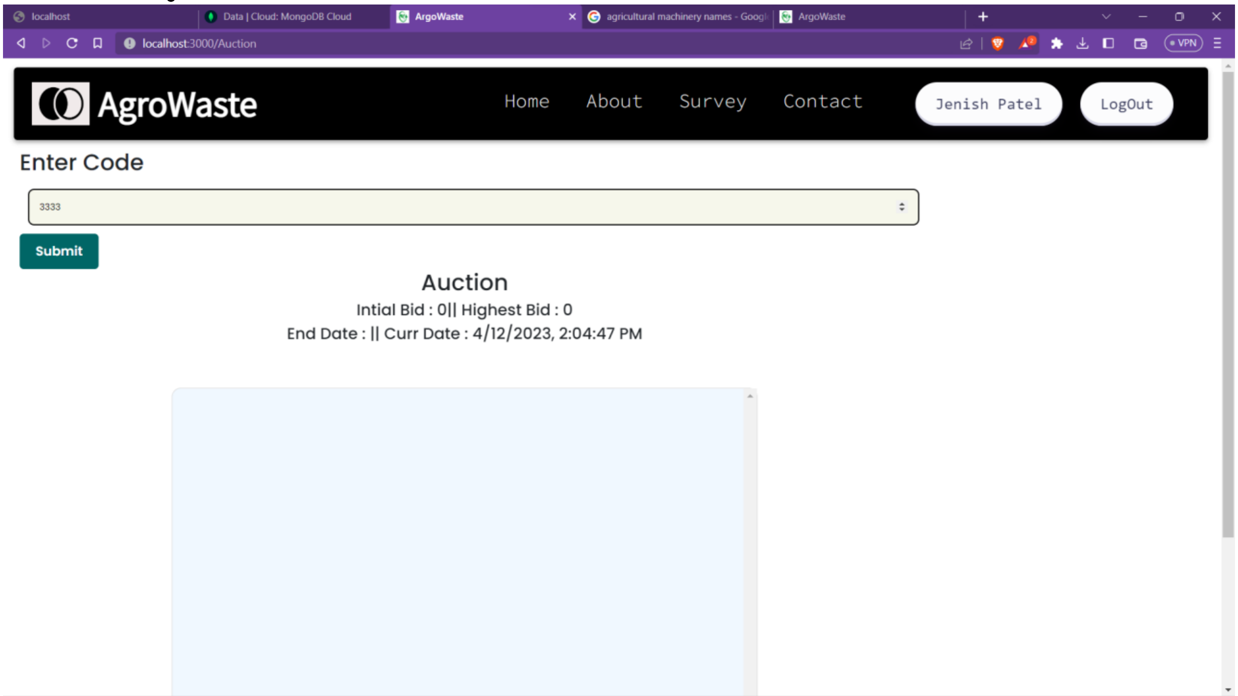


Fig 8.23 Unique Code Entry

- E

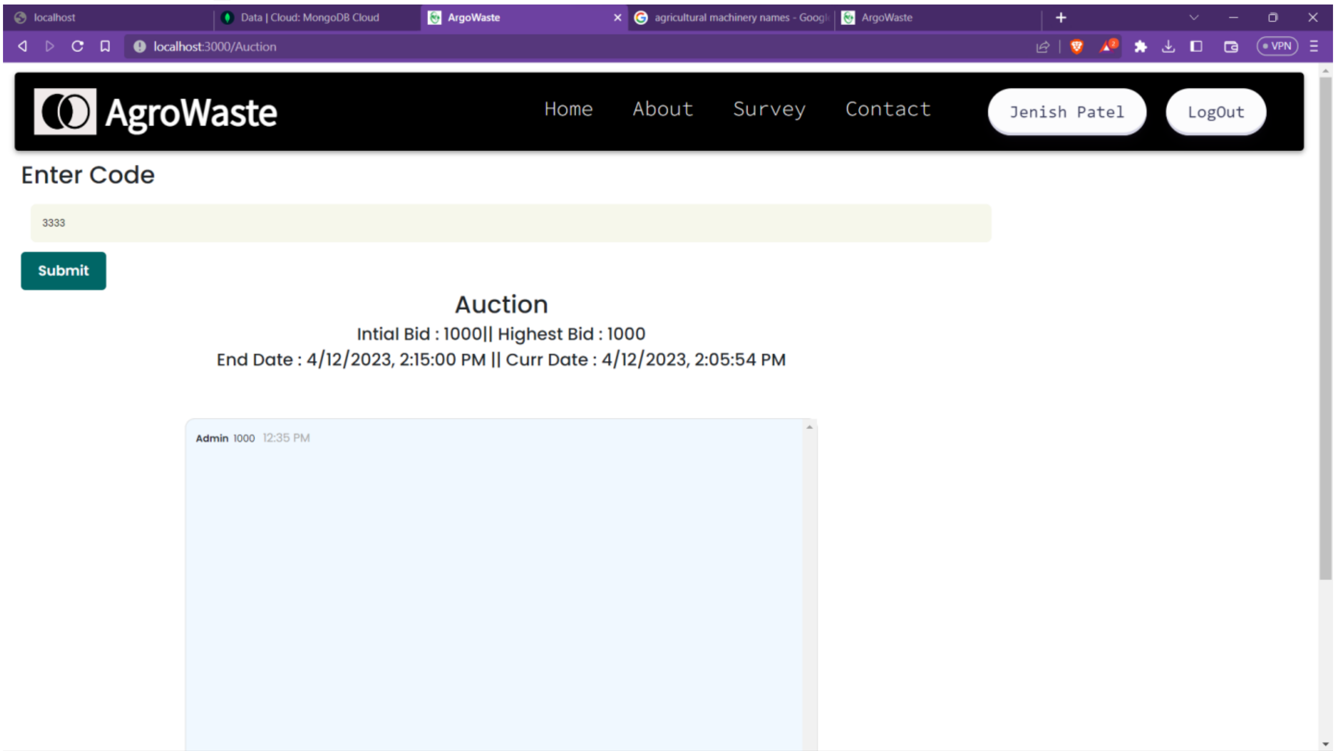


fig 8.24 Entering into Auction

- Entering Bid

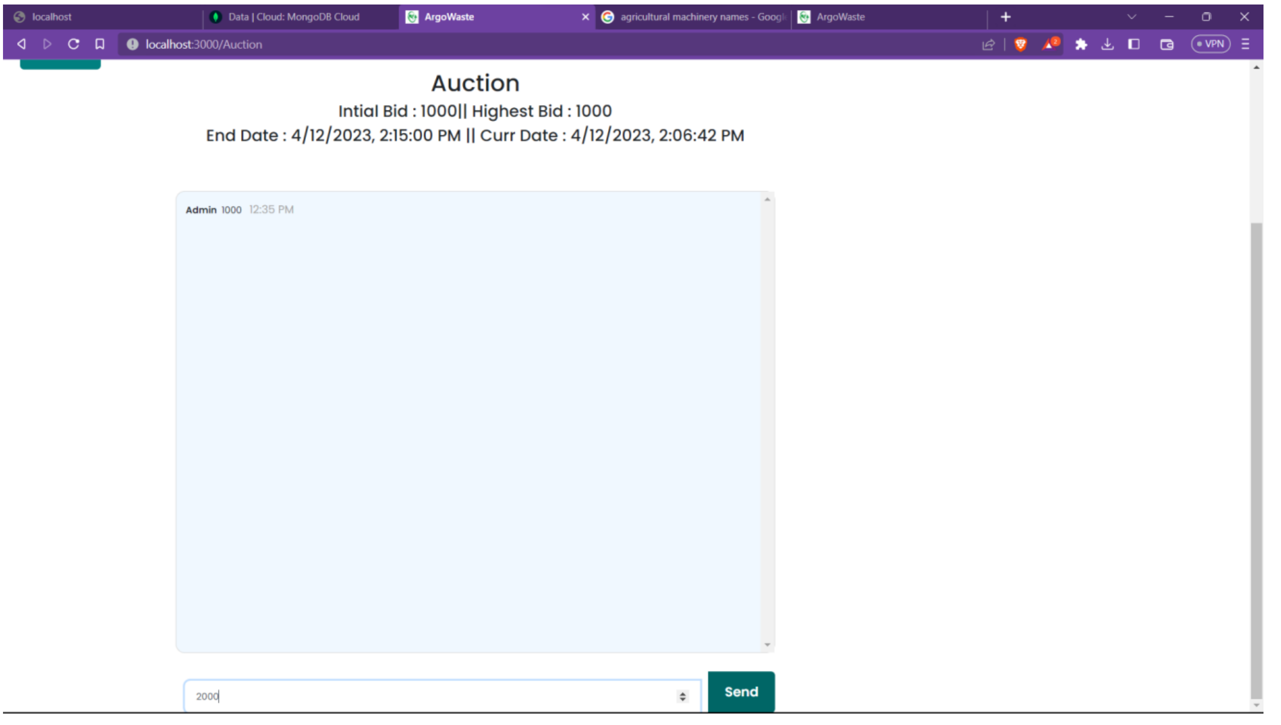


Fig 8.25 Entering bid

- Submit Bid by Jenish

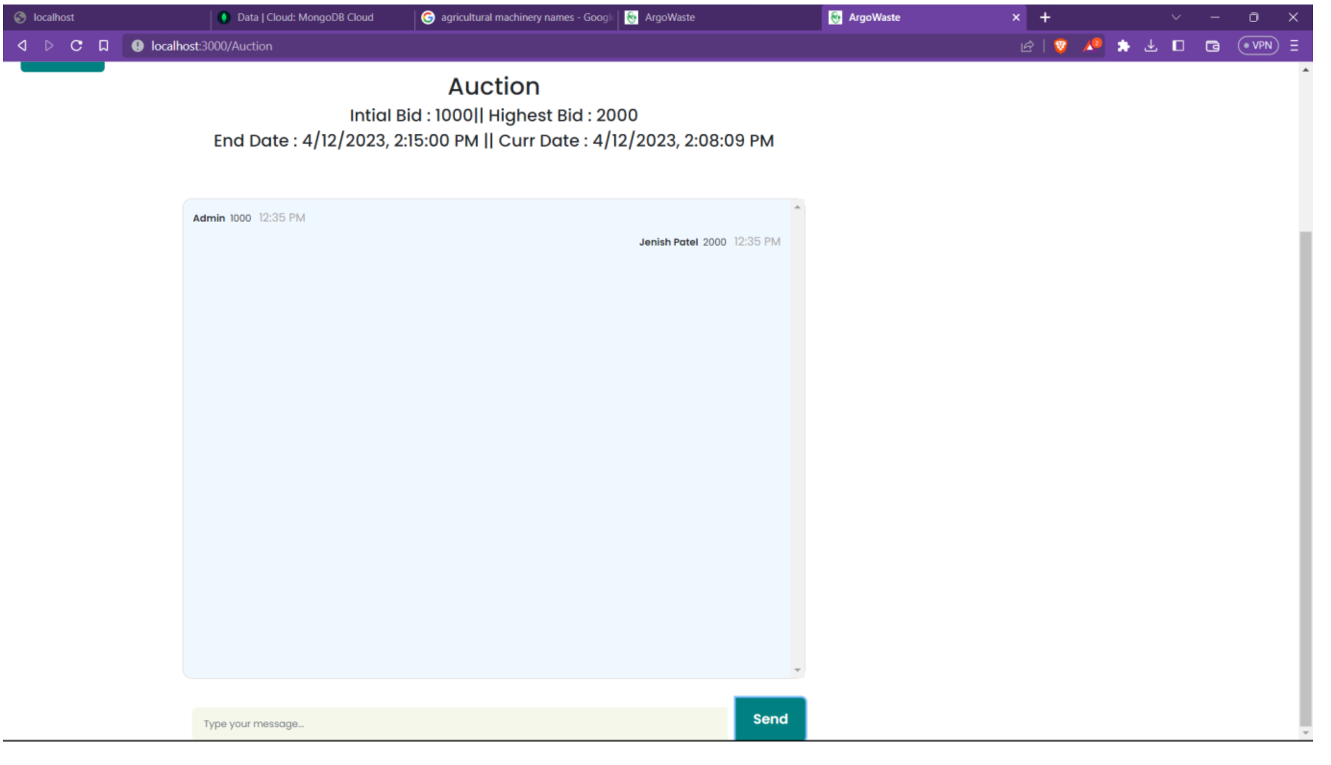


Fig 8.26 Jenish Entered the Bid

- Submit Bid by Gita (Another Buyer)

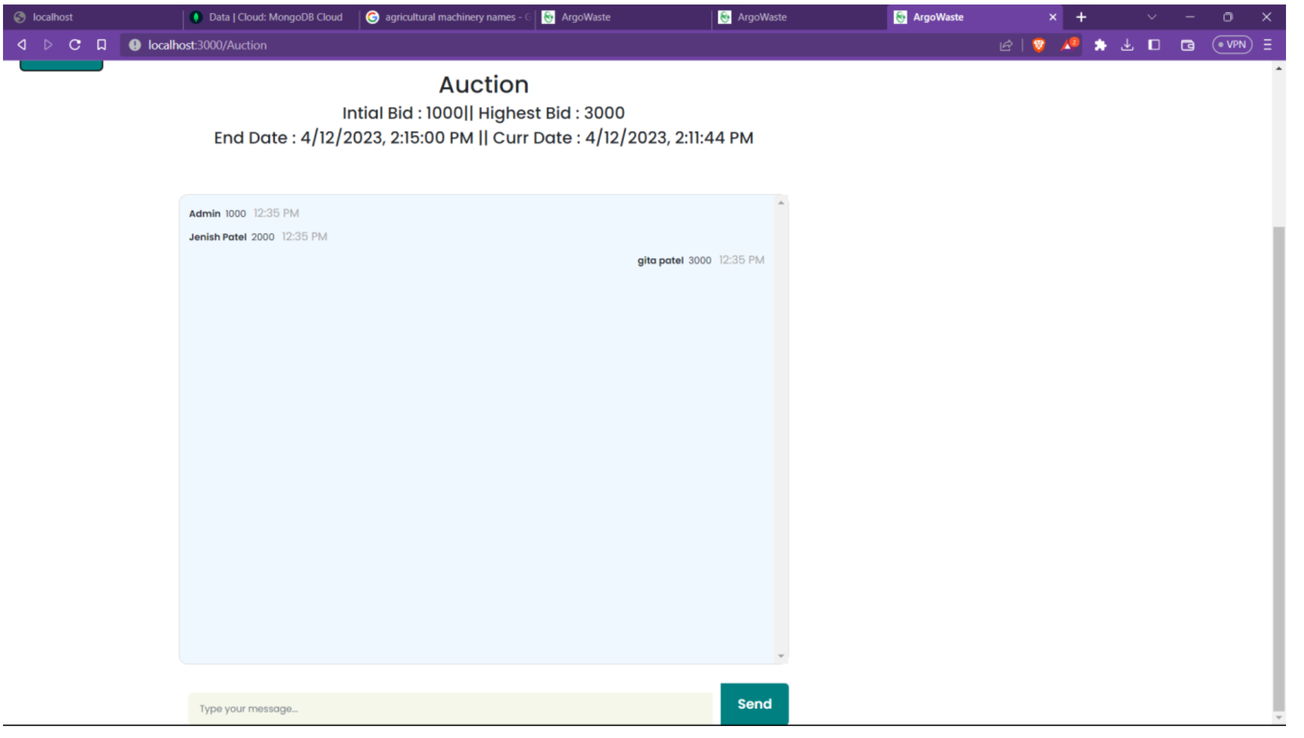


Fig 8.27 Bid by Gita

- Result Announced after End Date is over.

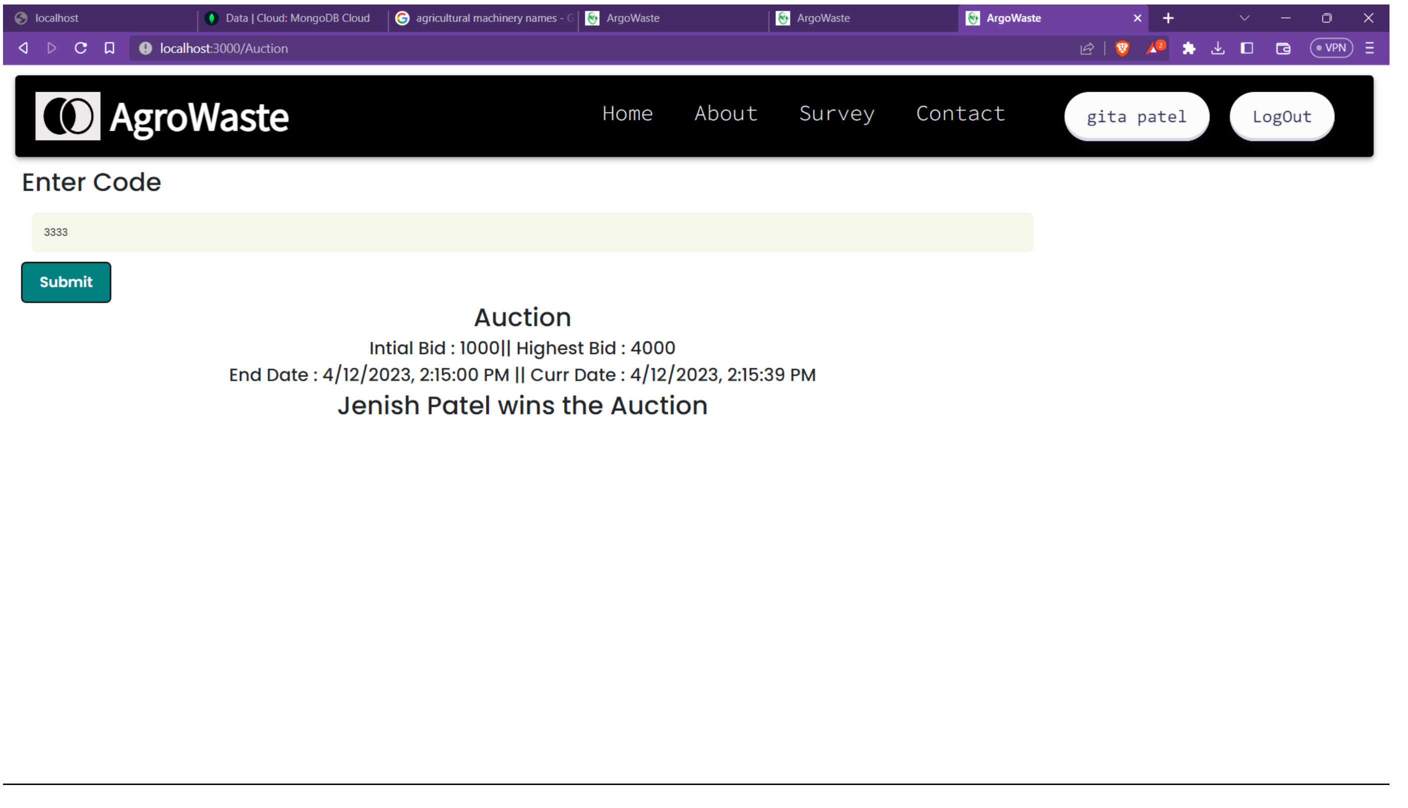


Fig 8.28 Auction Result

Admin Panel

1. Registration/Log In

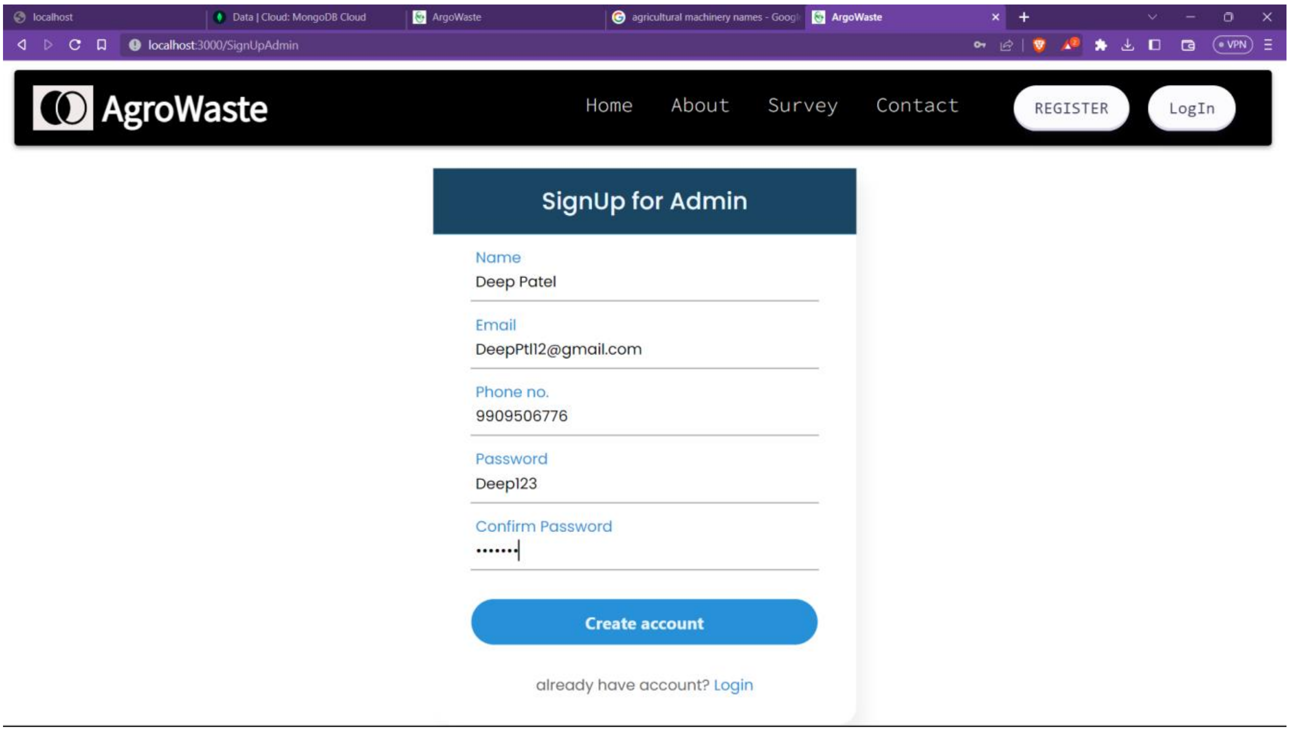


Fig 8.29 Admin sign up

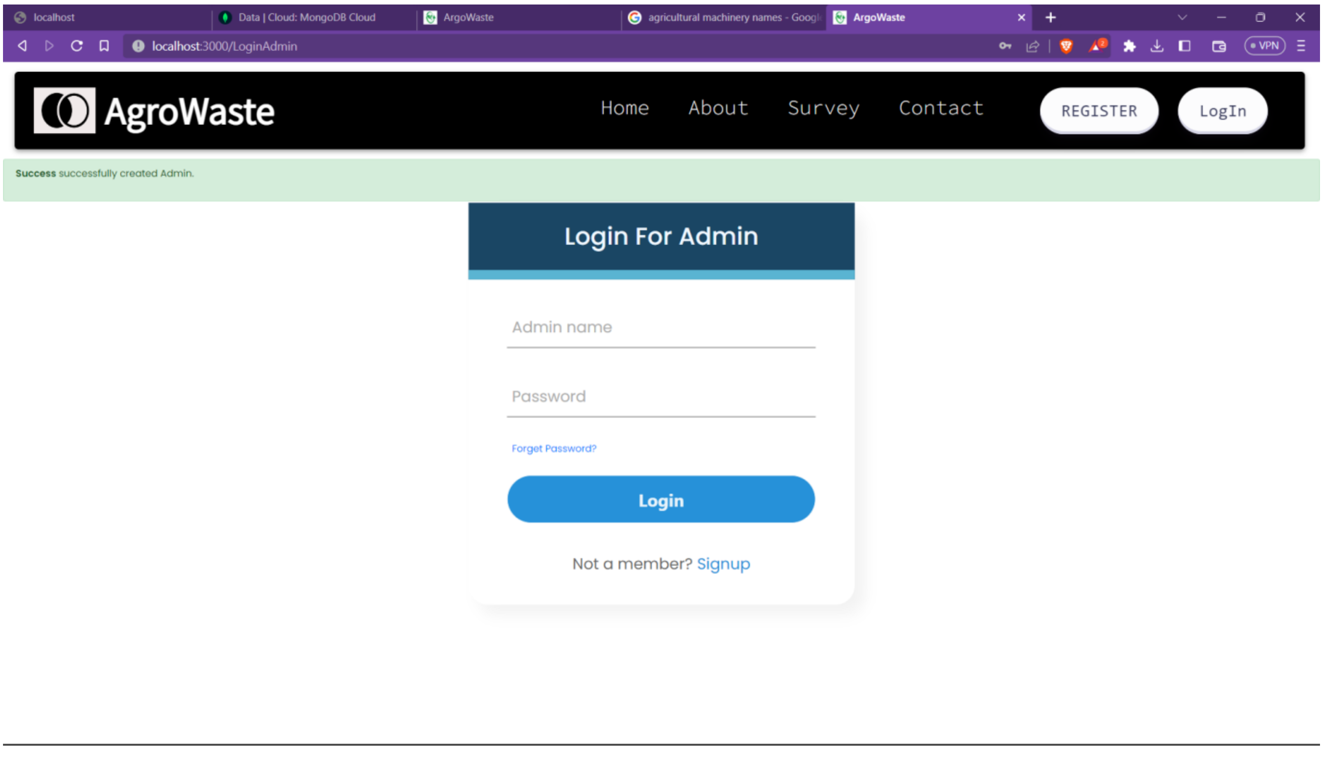


Fig 8.30 Admin Log in redirected

- Login

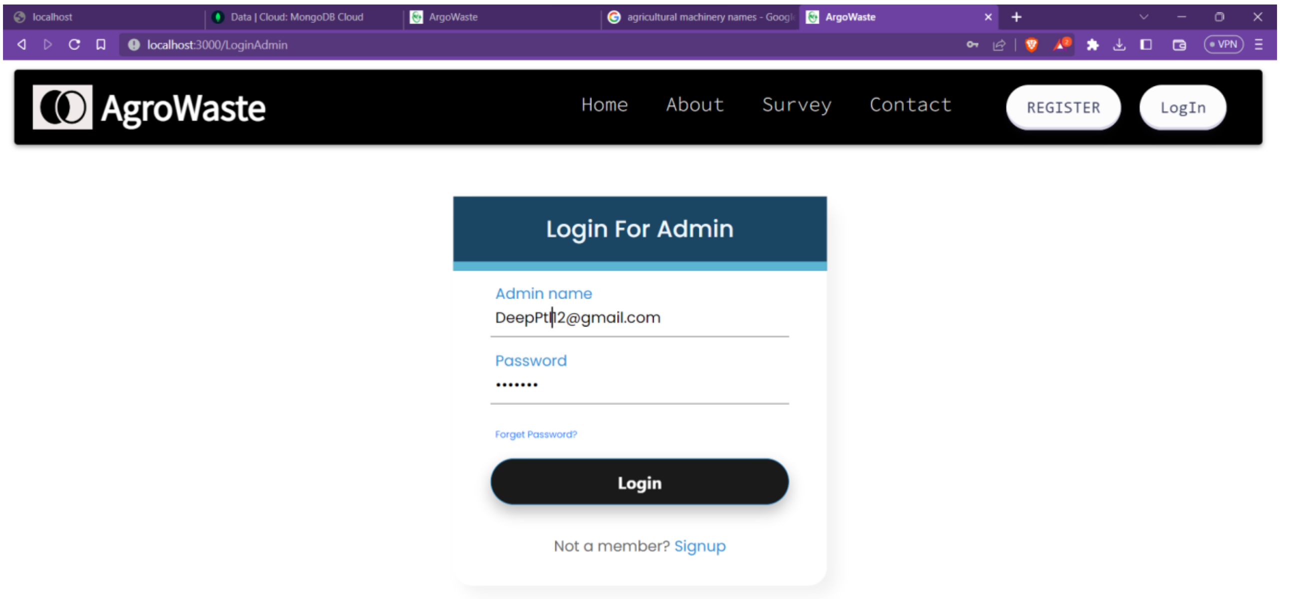


Fig 8.31 Admin Log in

- Admin Home page

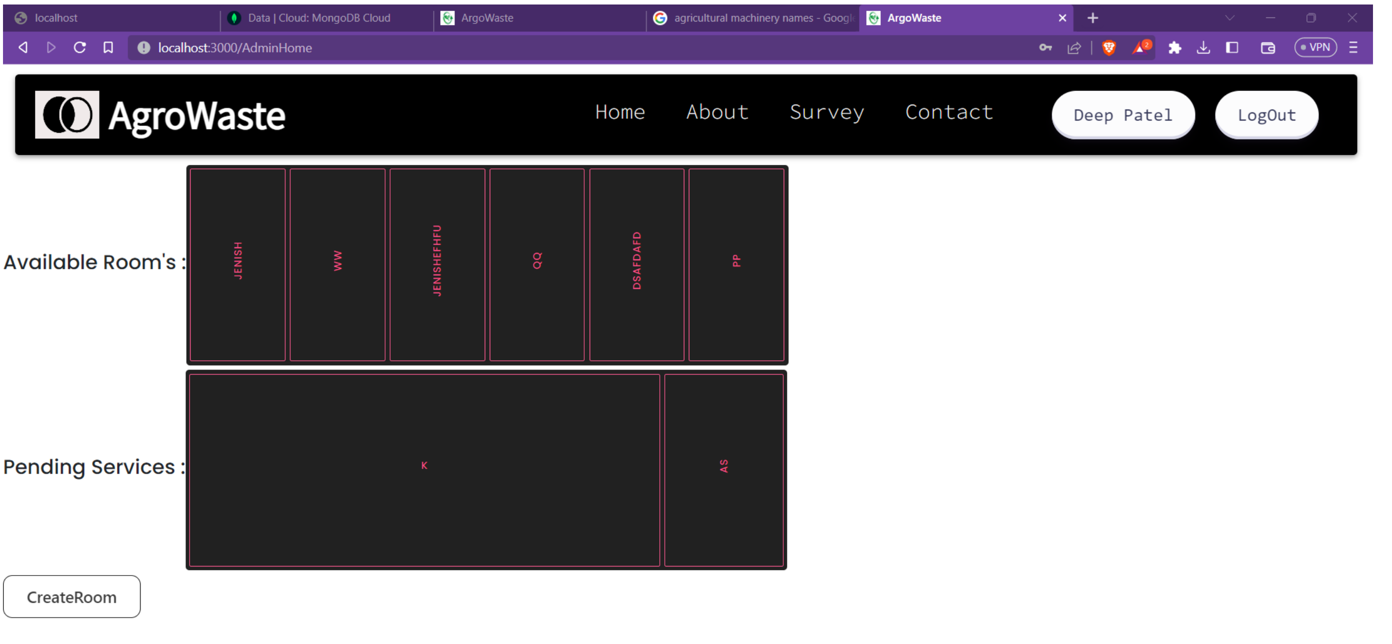


Fig 8.32 Admin Home page

2. Creating Room

- Create a new room for auction also mention end date and unique code.

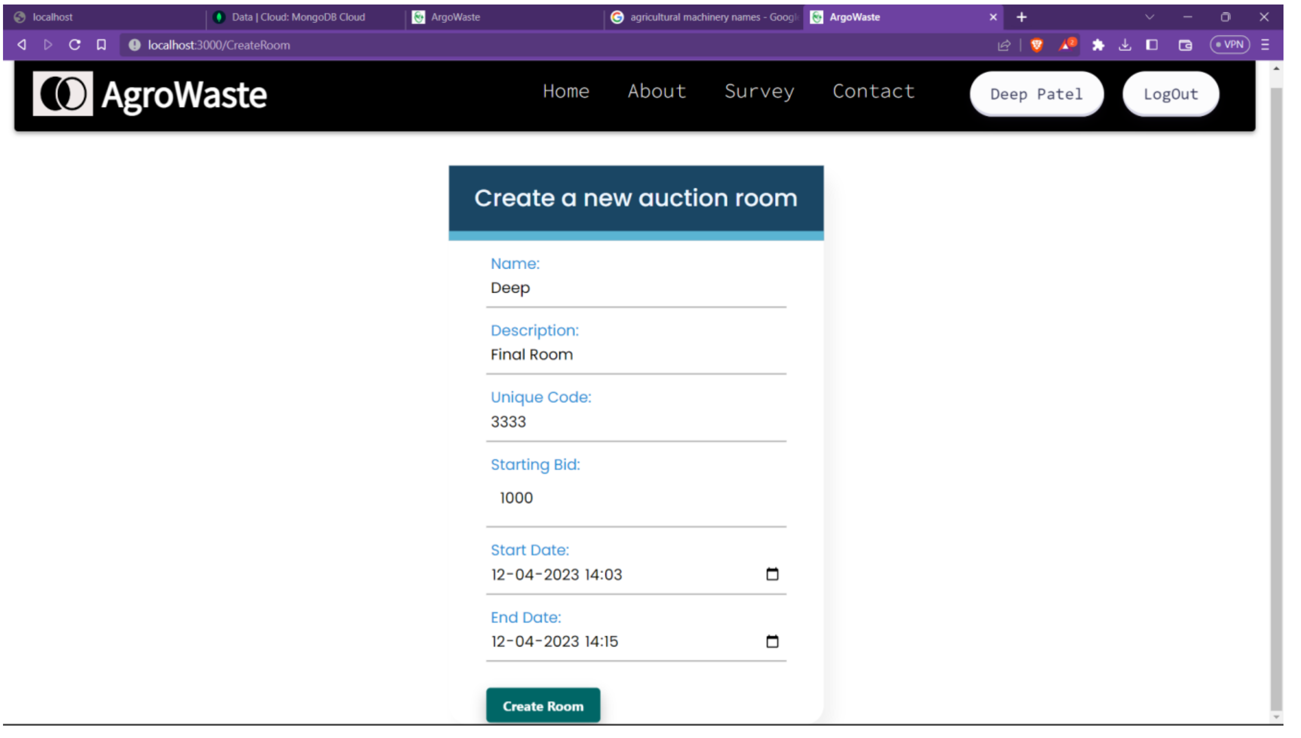


Fig 8.33 Creating a new room for Auction

- Created room was added to Home page.

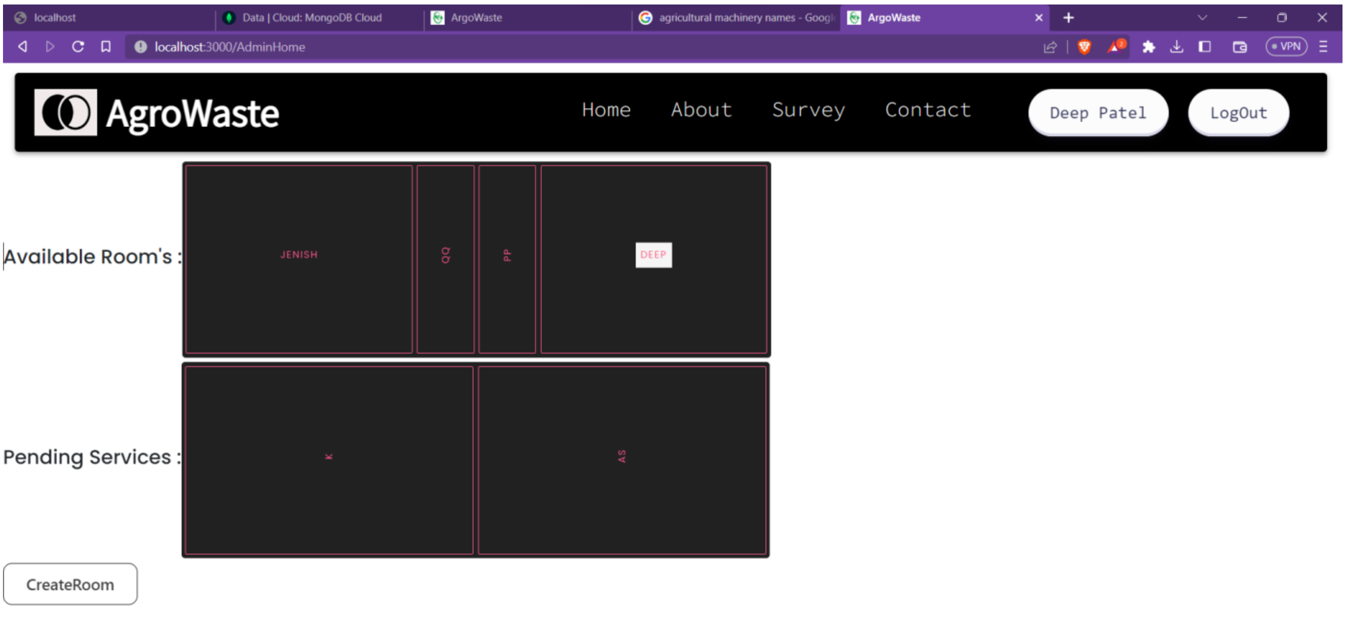


Fig 8.34 Room add to home page

- Rooms details page

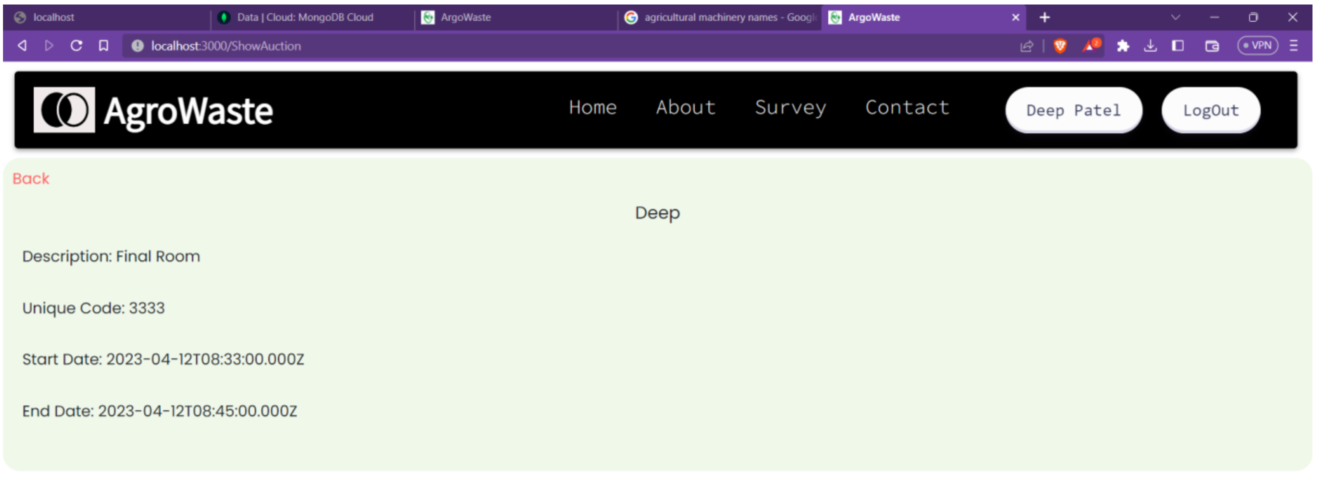


Fig 8.35 Rooms details

3. Managing incoming requests

- Grant/Approve incoming requests by farmers page through Pending Request List.

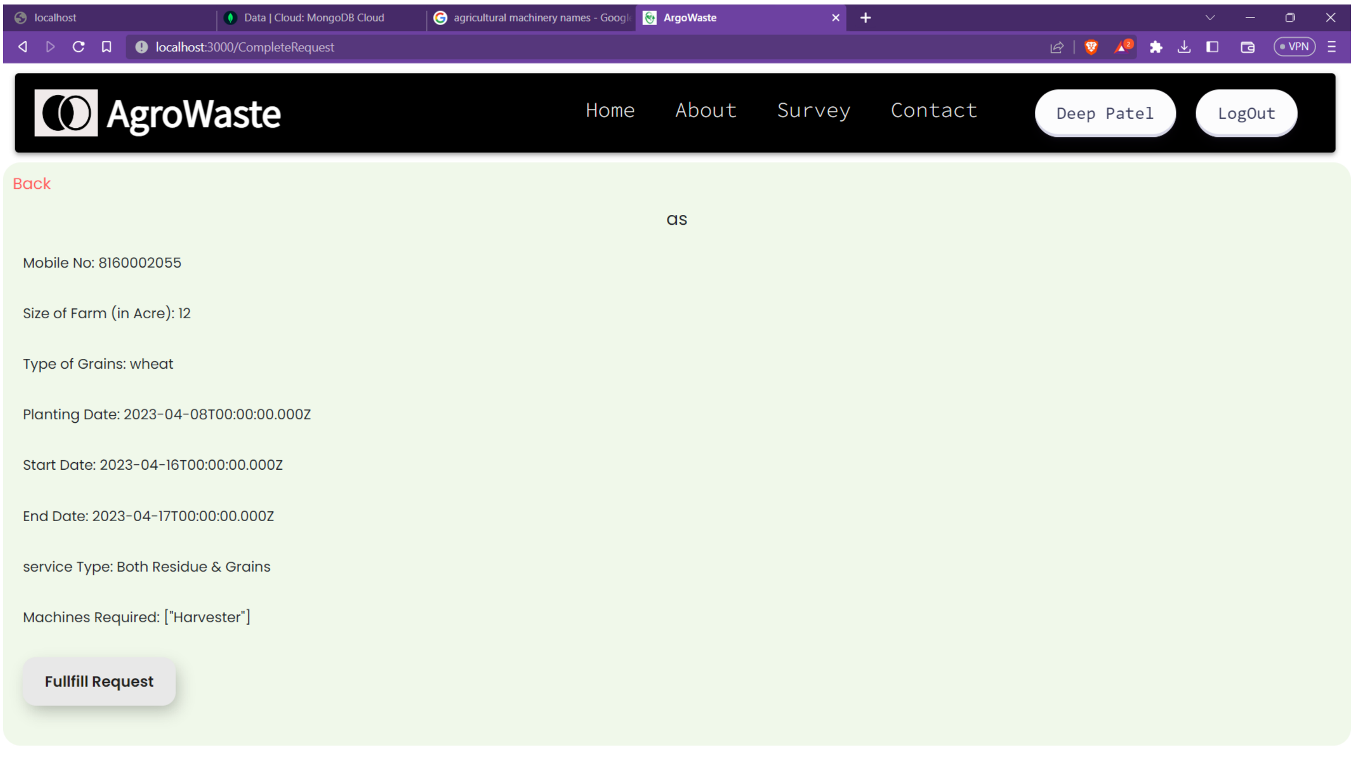


Fig 8.36 Approve farmer's request page

- After service completion, Fill a form and adding the data related to residue collection.

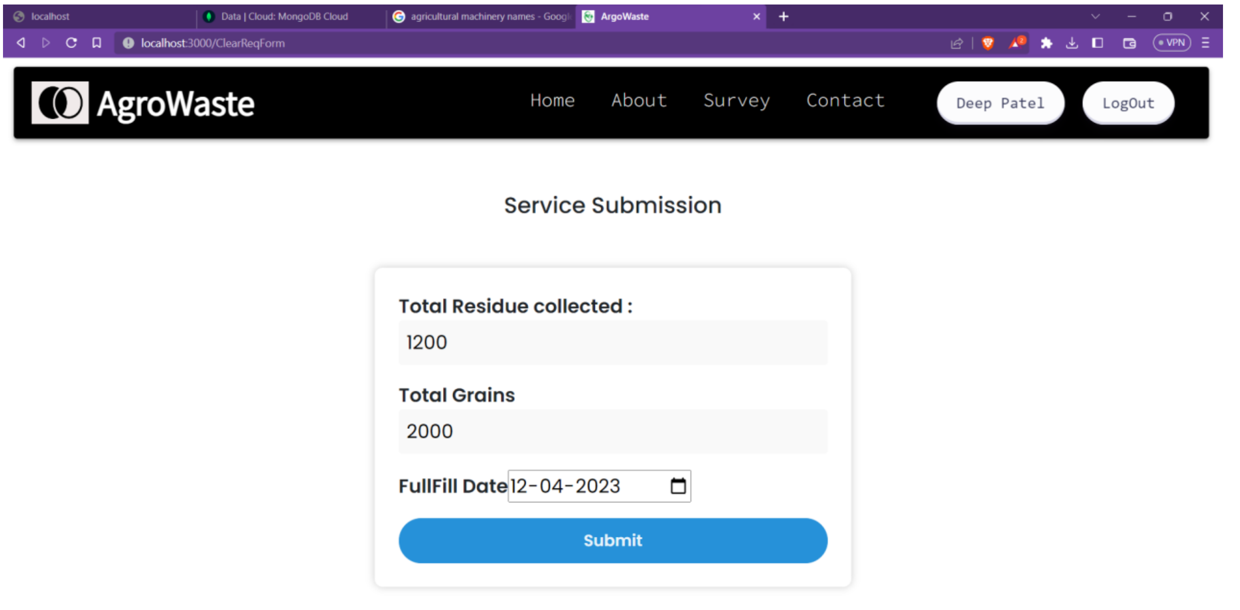


Fig 8.37 Service Submission

8.2 Classification of satellite image

Collection of Data

- Open a browser then serch for https://bhuvan.nrsc.gov.in
- Click on Open Data Archive
- Select the satellite sensor as given below

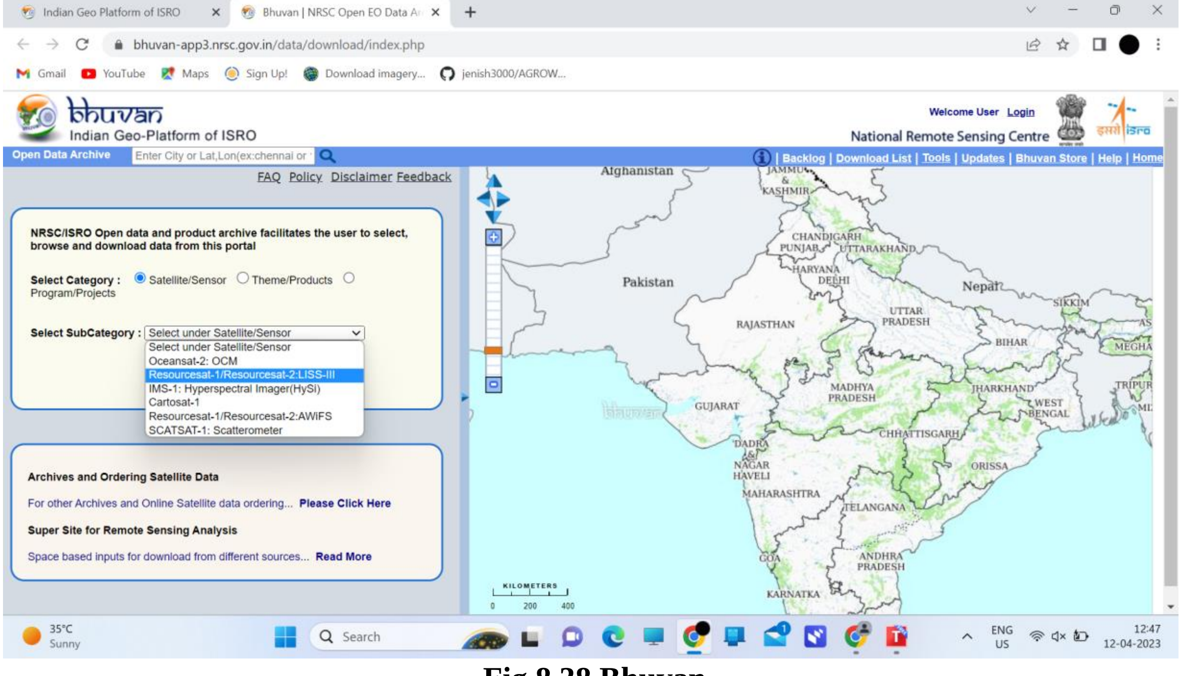


Fig 8.38 Bhuvan

- Select the desired area

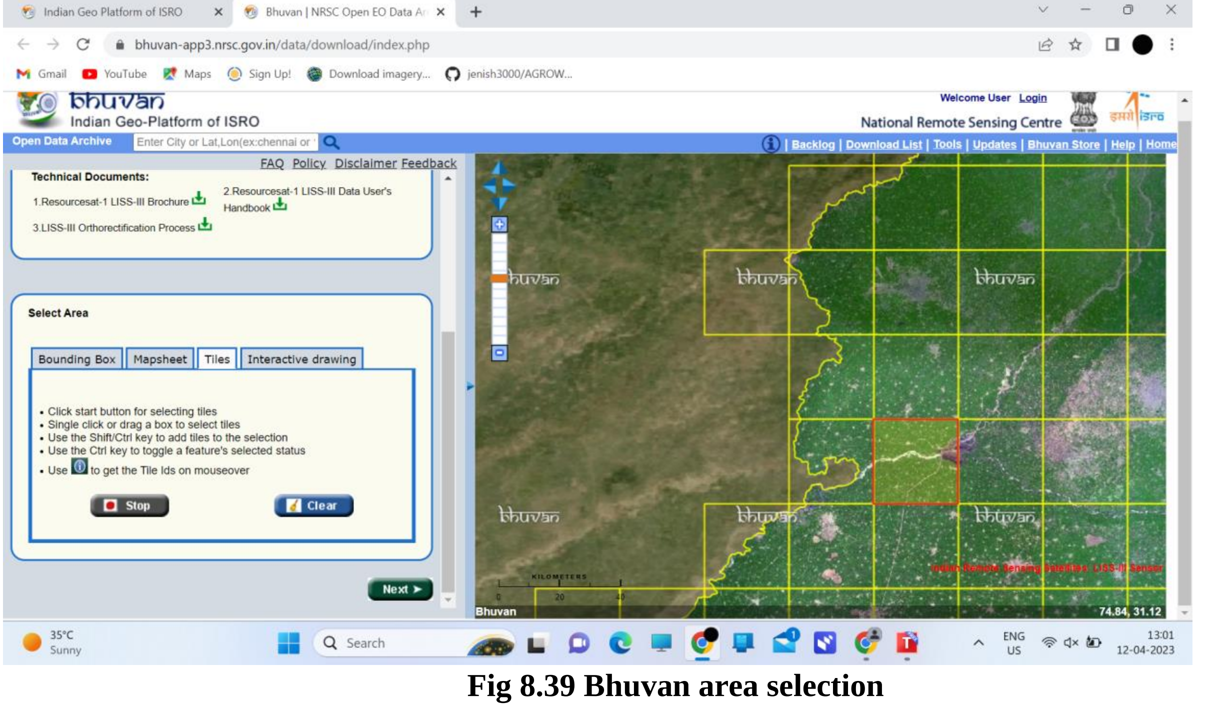


Fig 8.39 Bhuvan area selection

- Download the appropriate data .

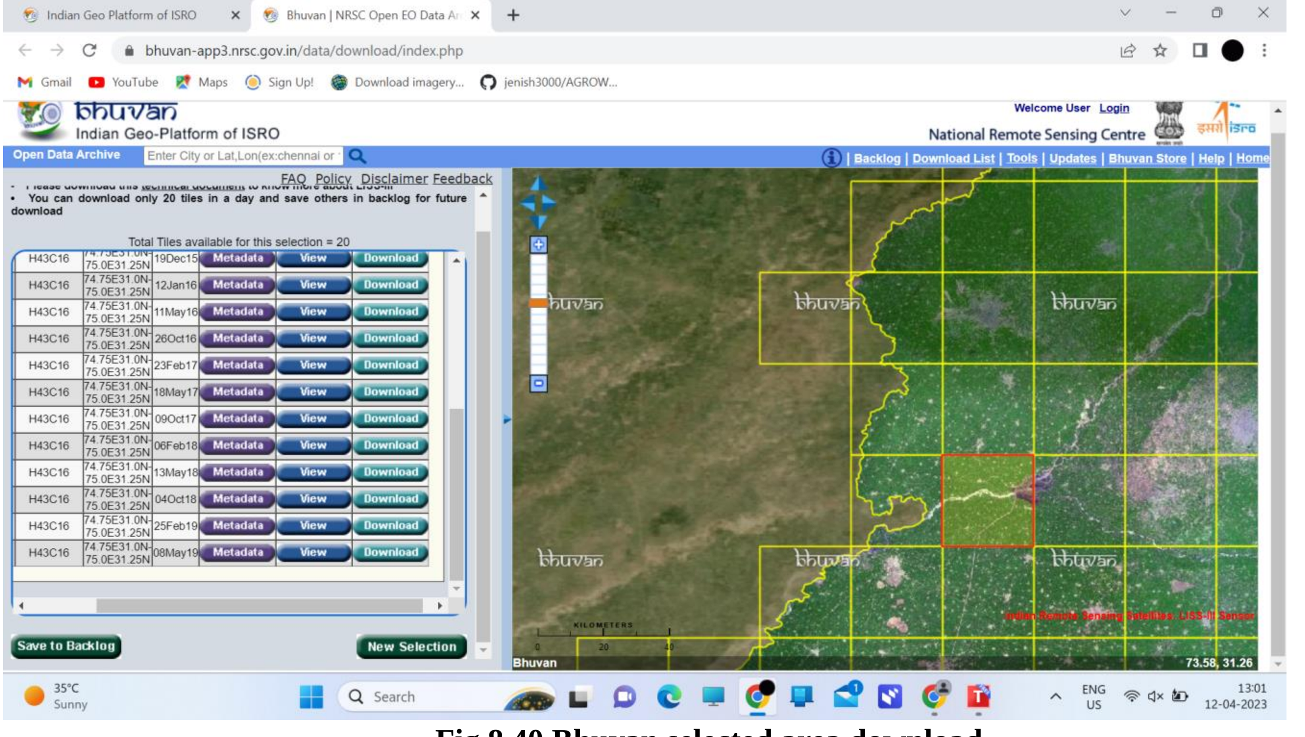


Fig 8.40 Bhuvan selected area download

- we can see that images are separated into to four bands

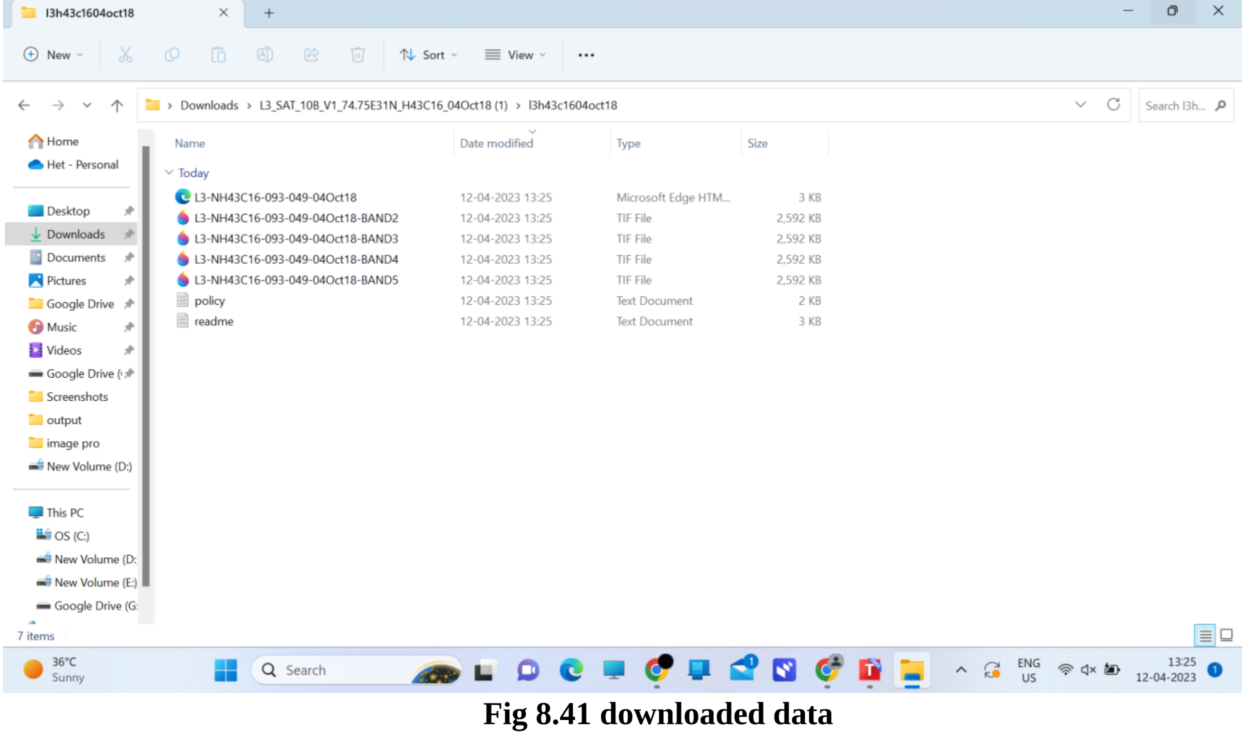


Fig 8.41 downloaded data

QGIS

- Download QGIS from https://www.qgis.org
- Open QGIS
- Install Semi automatic classification for better visualization
- Add raster layer

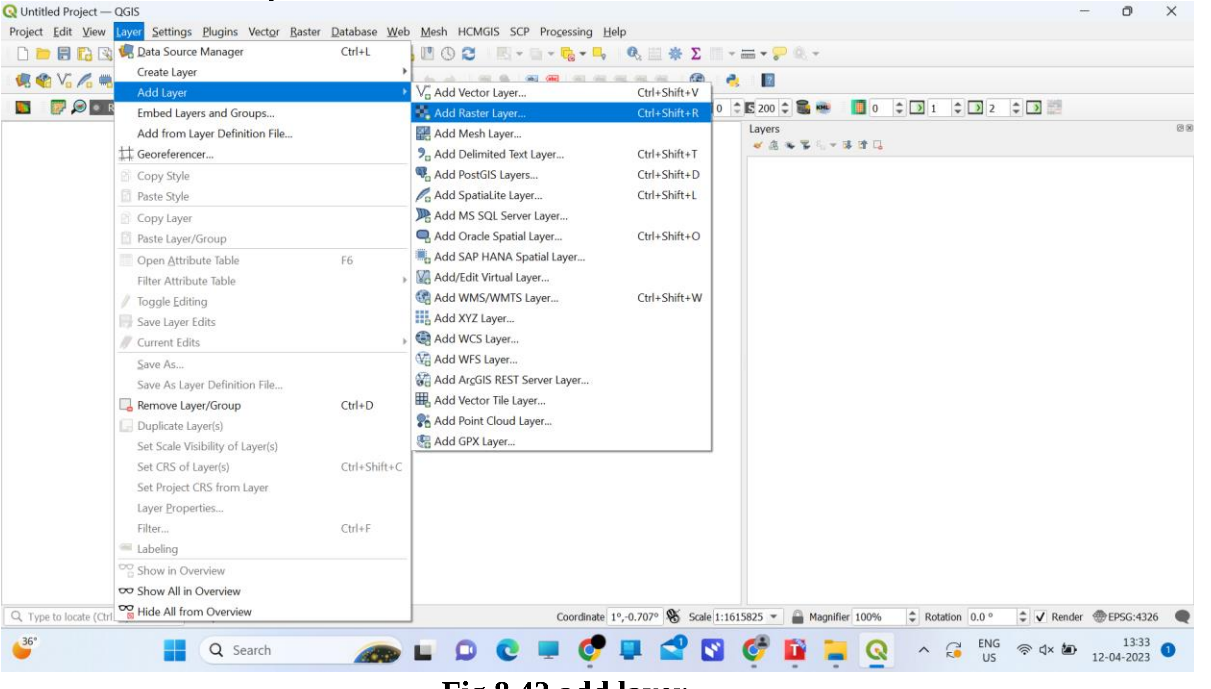


Fig 8.42 add layer

- select four tif files which are downloaded and add those layers.

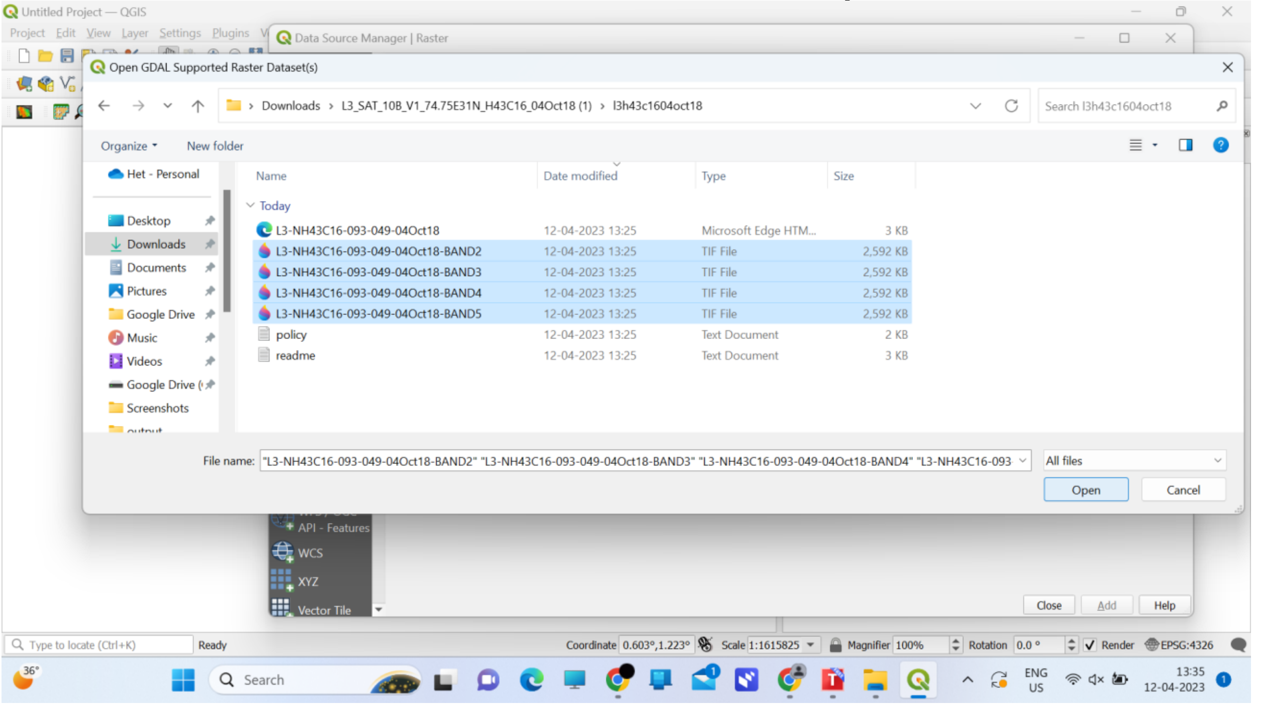


Fig 8.43 layer selection

- Apply semi automatic classification

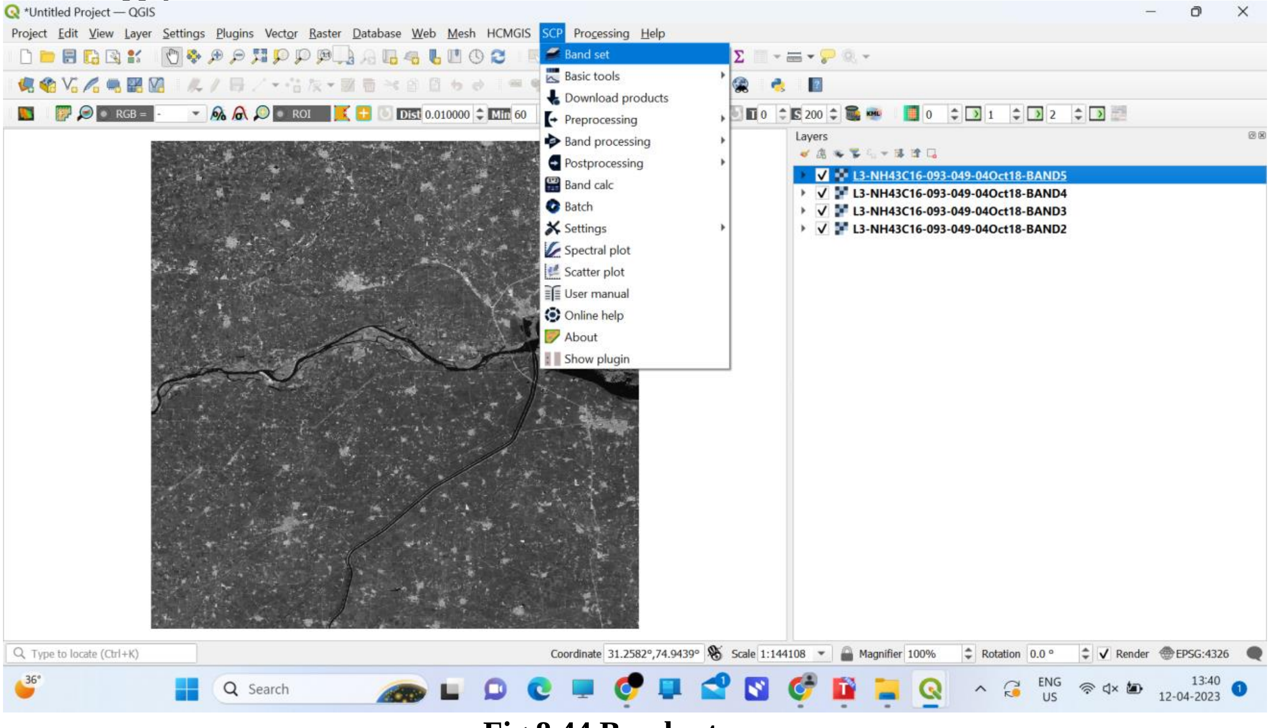


Fig 8.44 Band set

- set the band combination and create raster of band set

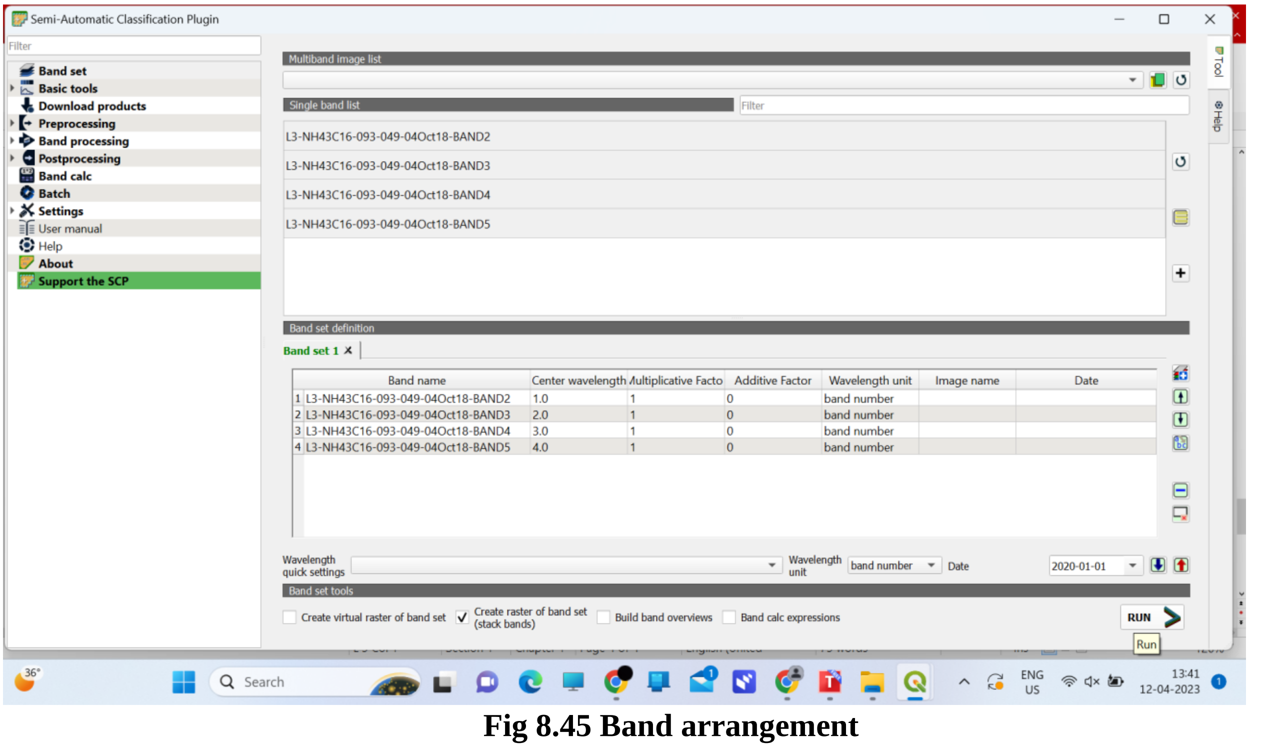


Fig 8.45 Band arrangement

- For training select show plugin option.

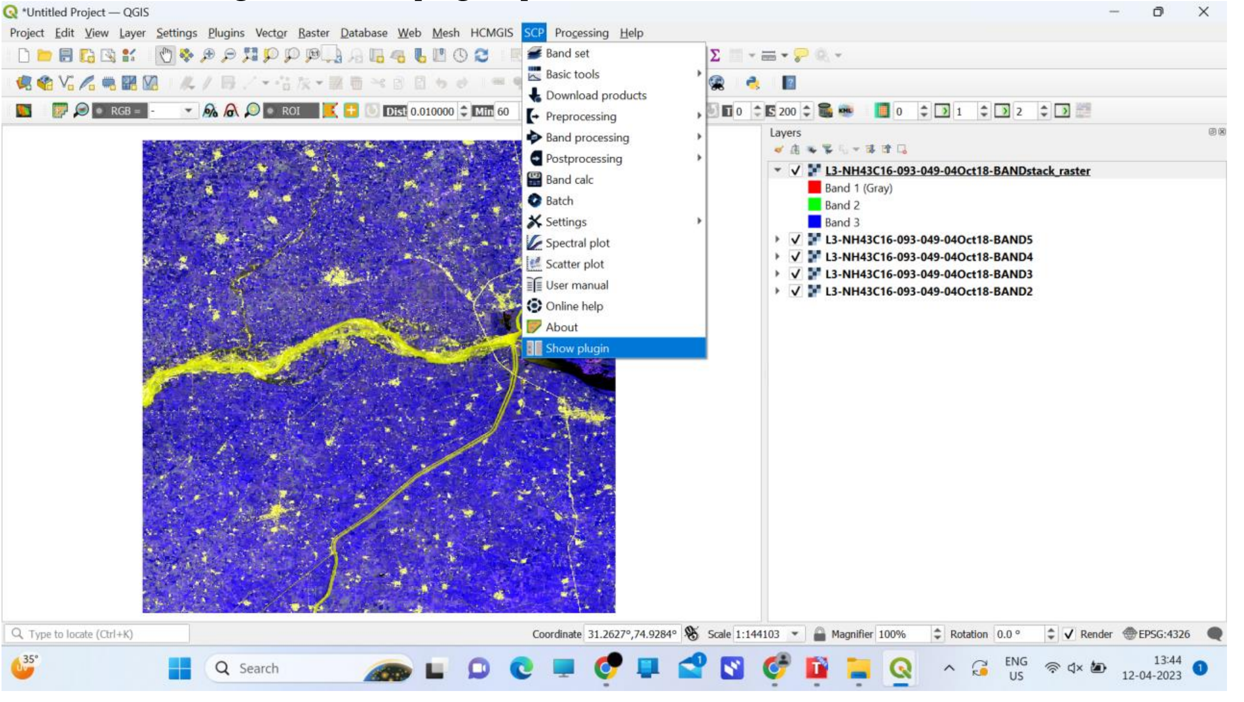


Fig 8.46 Training

- select Traning input and create the scp file

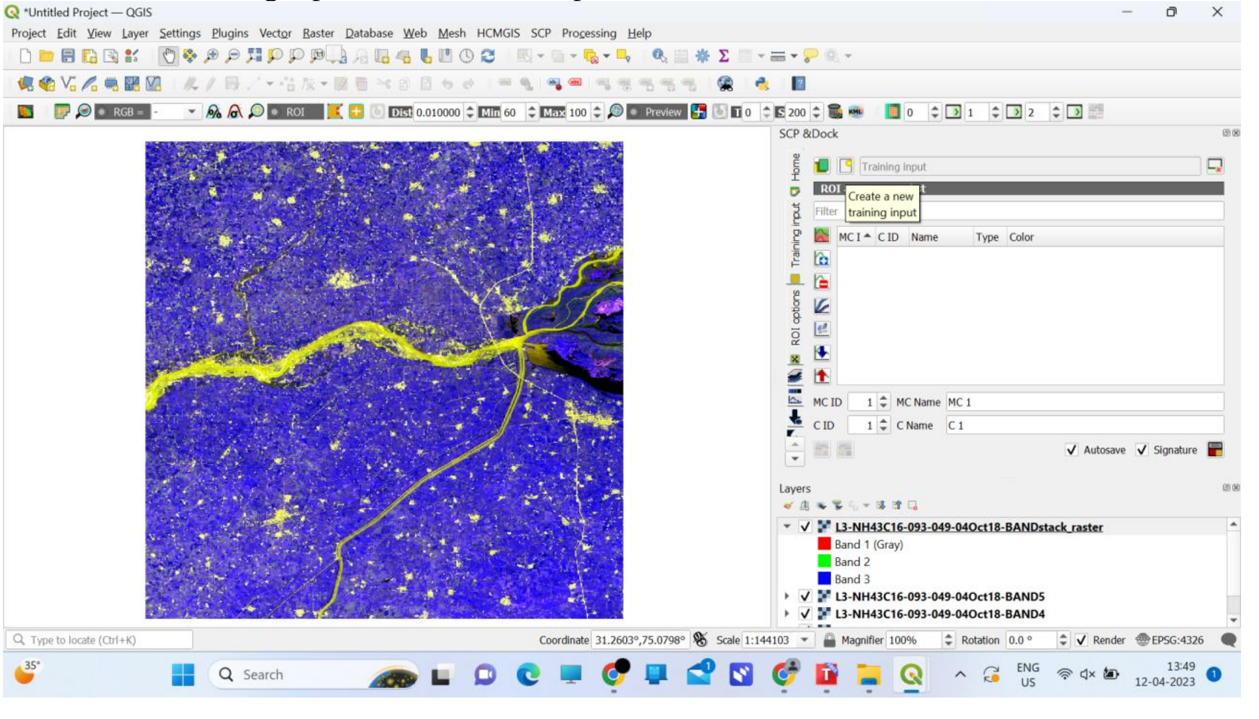


Fig 8.47 Create scp(Training data) file

- select the icon of polygon for creating polygon

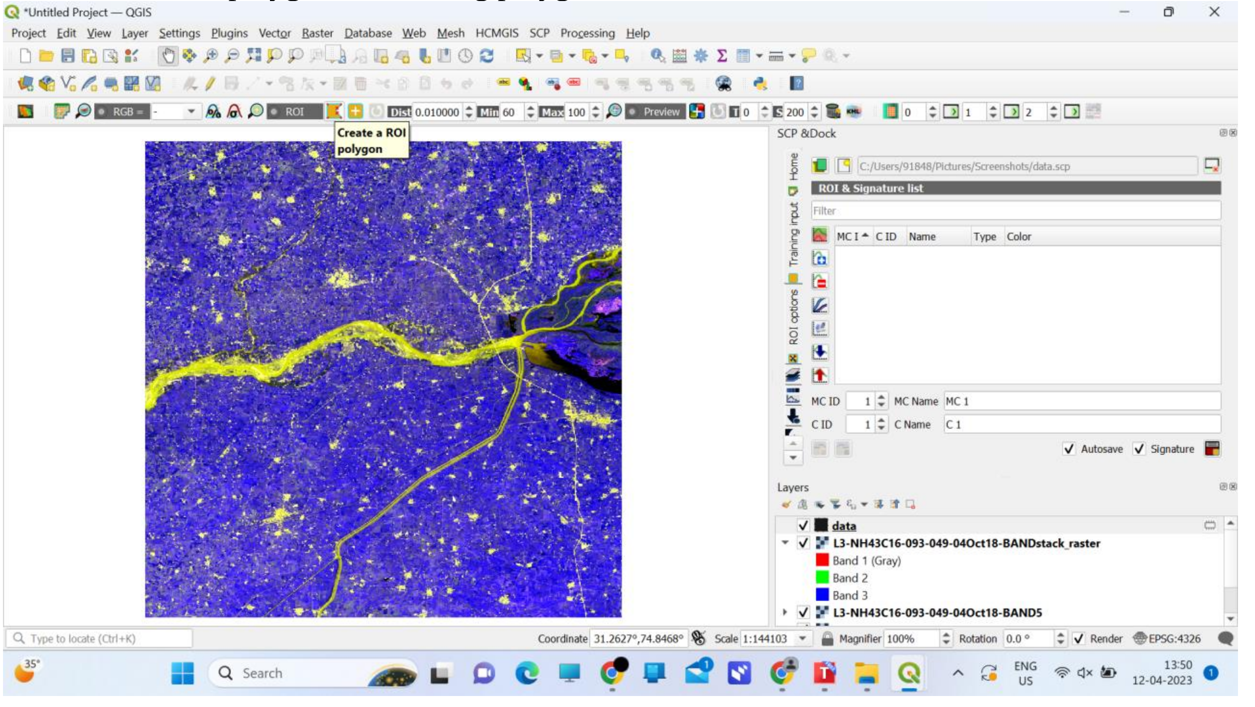


Fig 8.48 Input as a polygon

- Adding city input

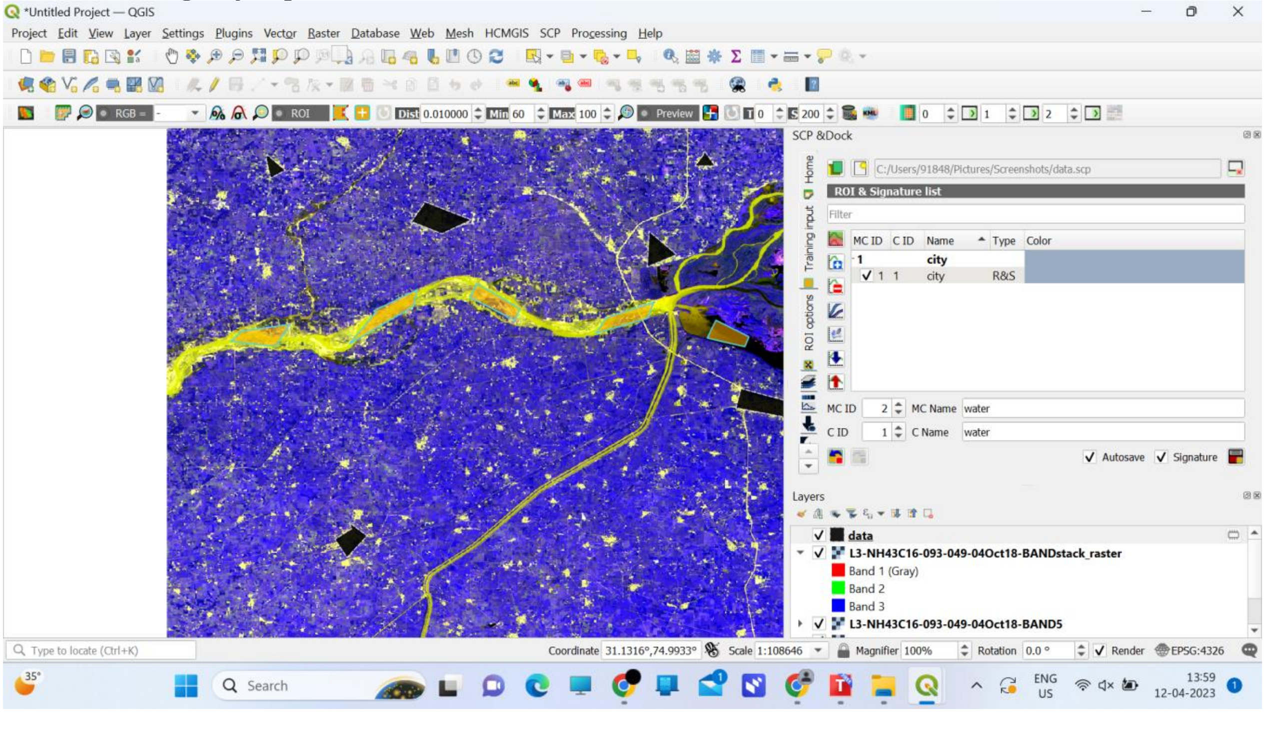


Fig 8.49 City Input as a training

- Adding water input

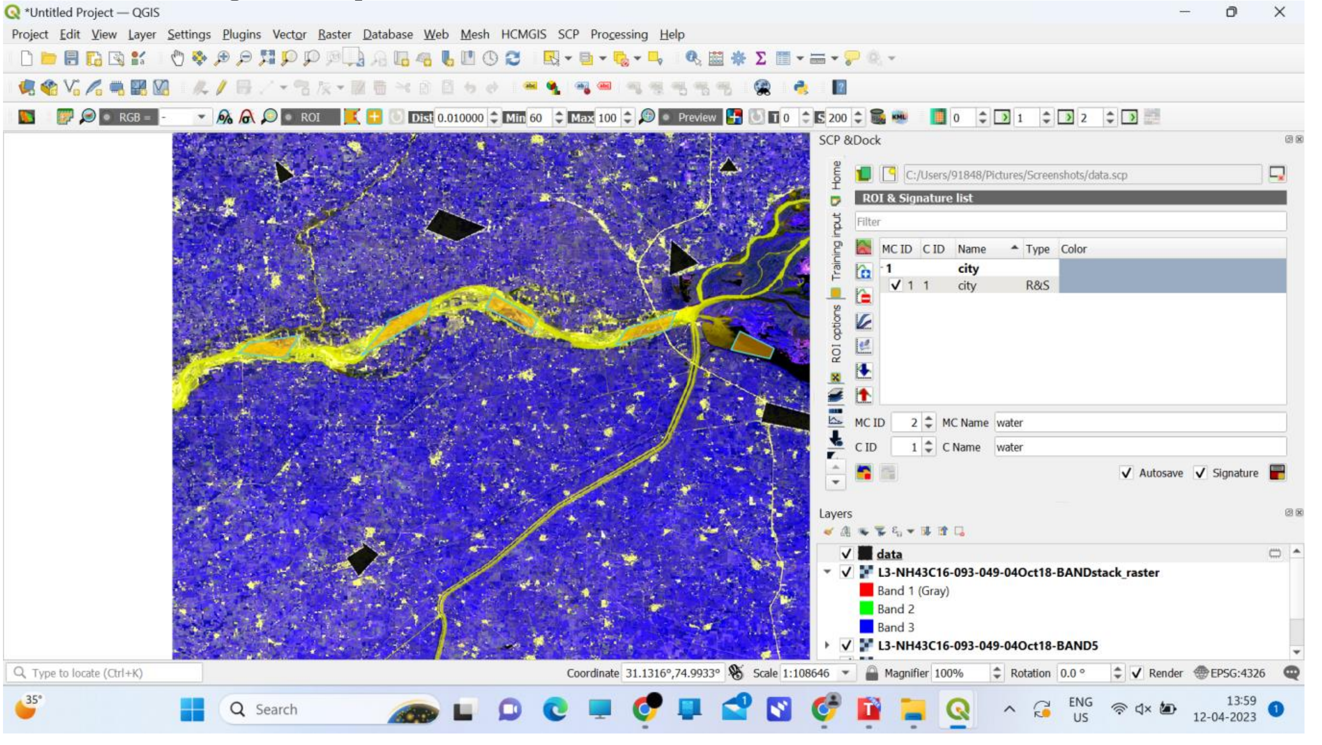


Fig 8.50 Water Input as a training

- Adding Farming green field input

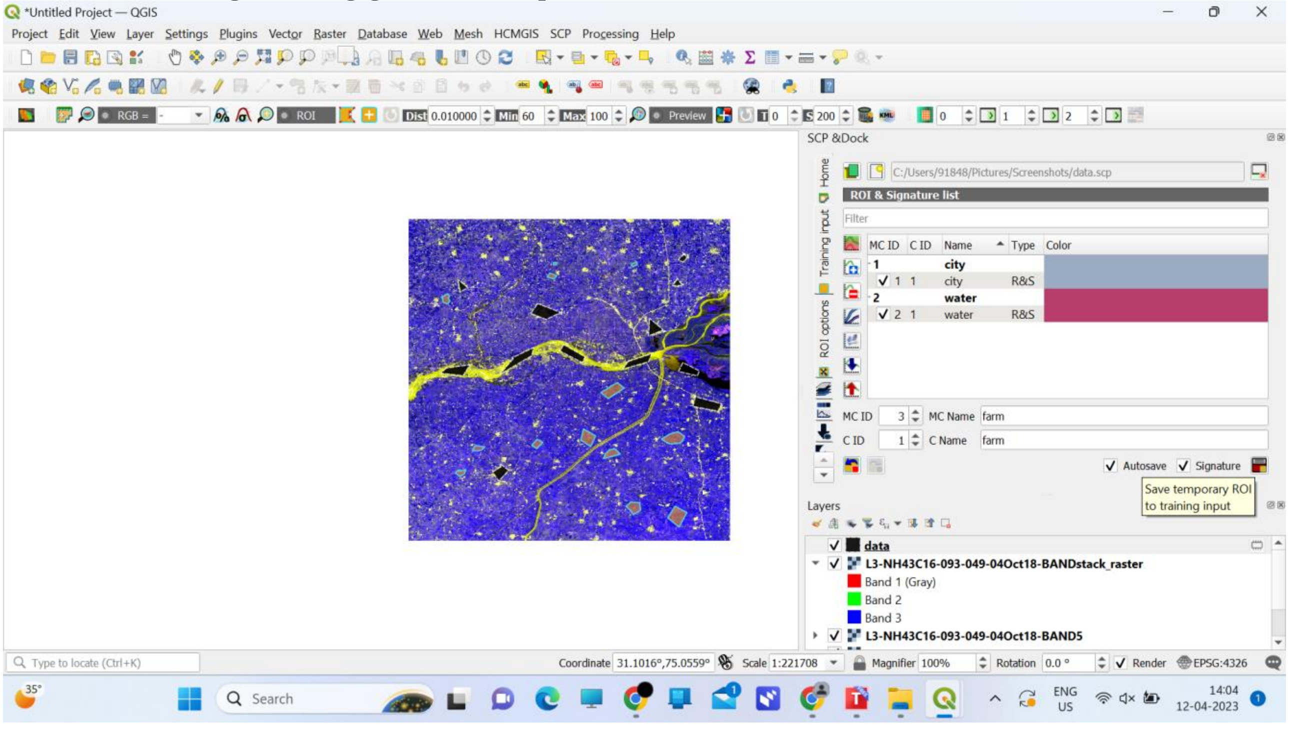


Fig 8.51 Farm Input as a training

- Adding red soil area input

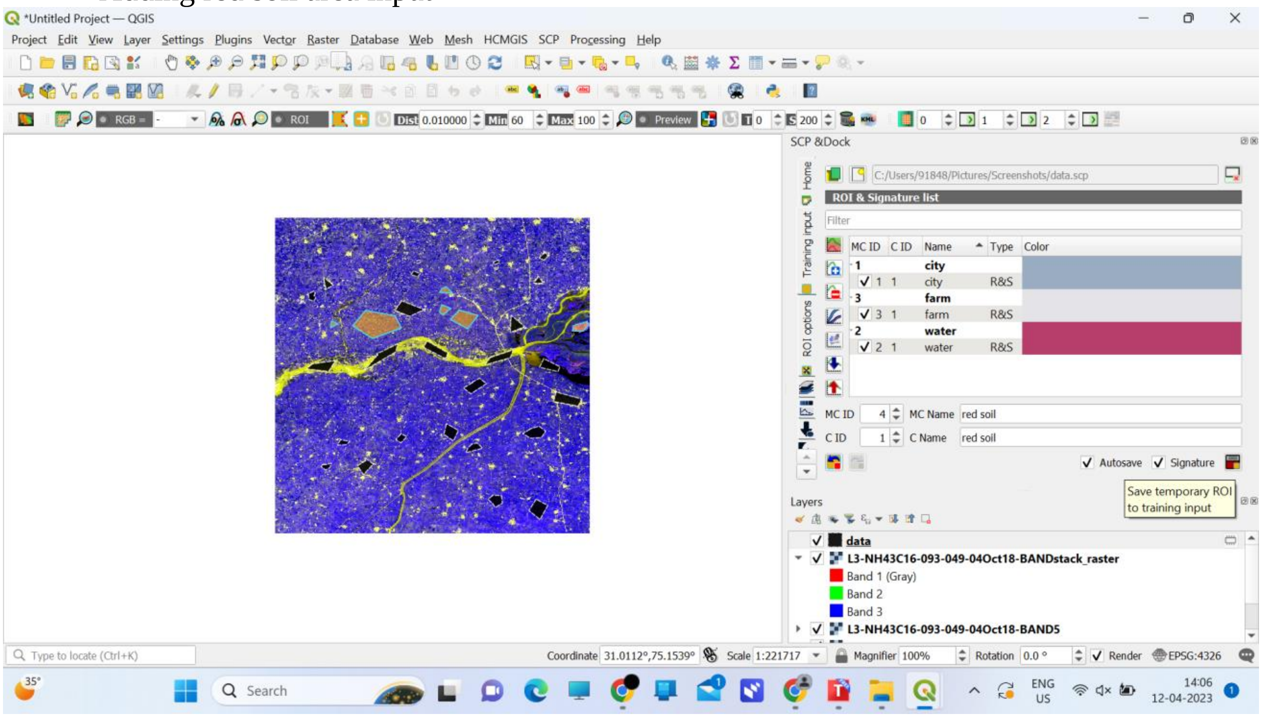


fig 8.52 Red soil Input as a training

- select Band processing and inside it select classification

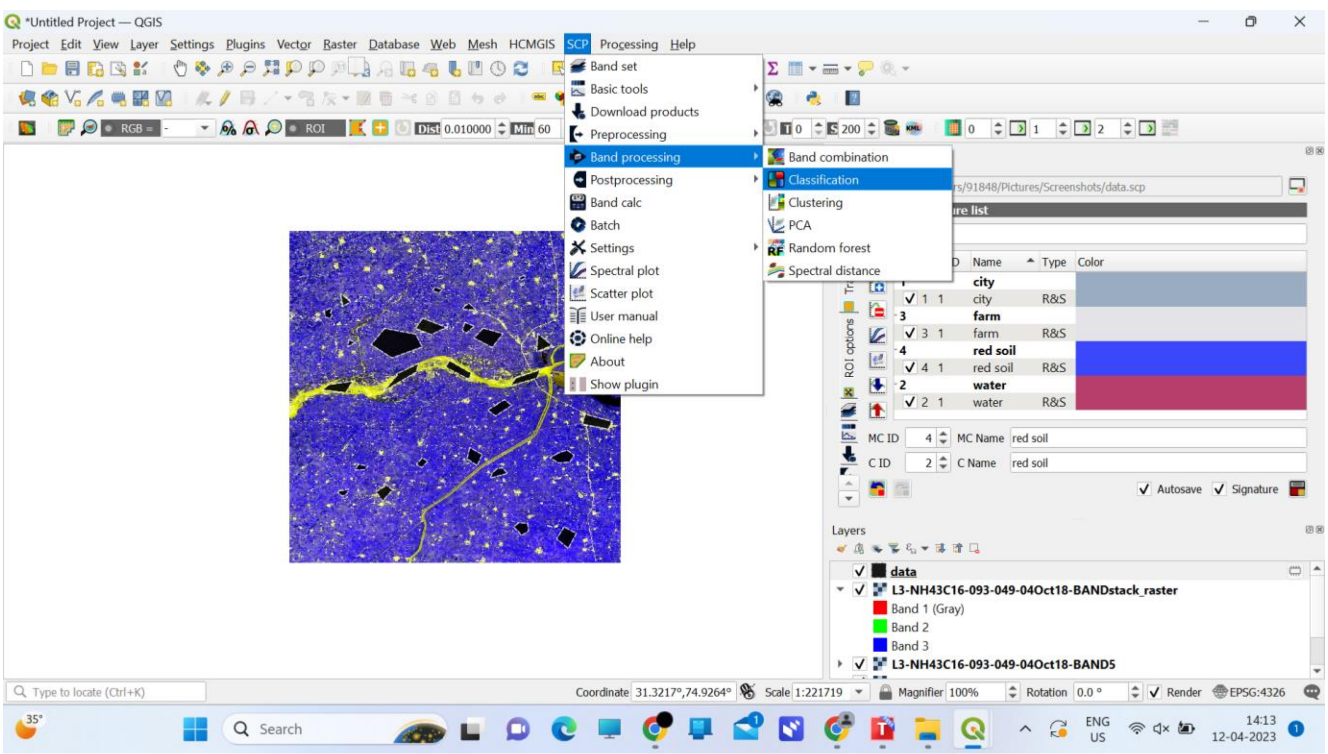


Fig 8.53 Classification

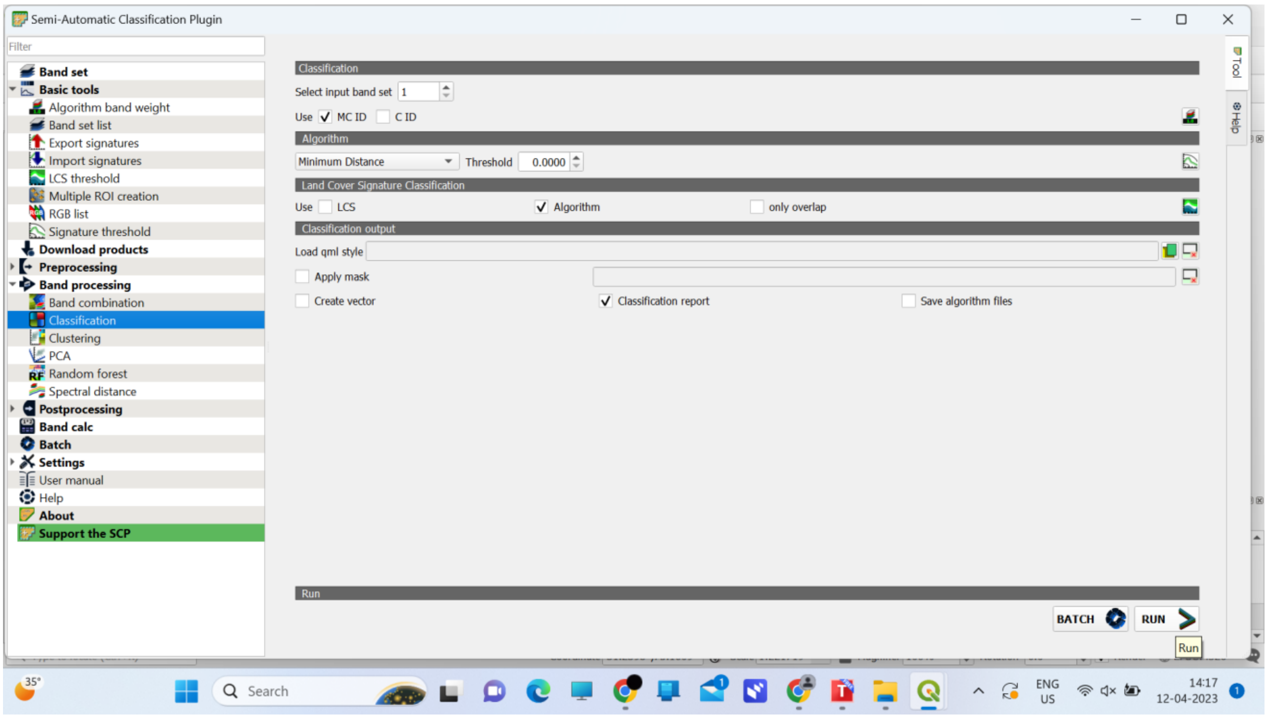


Fig 8.54 Classification selection

- Output

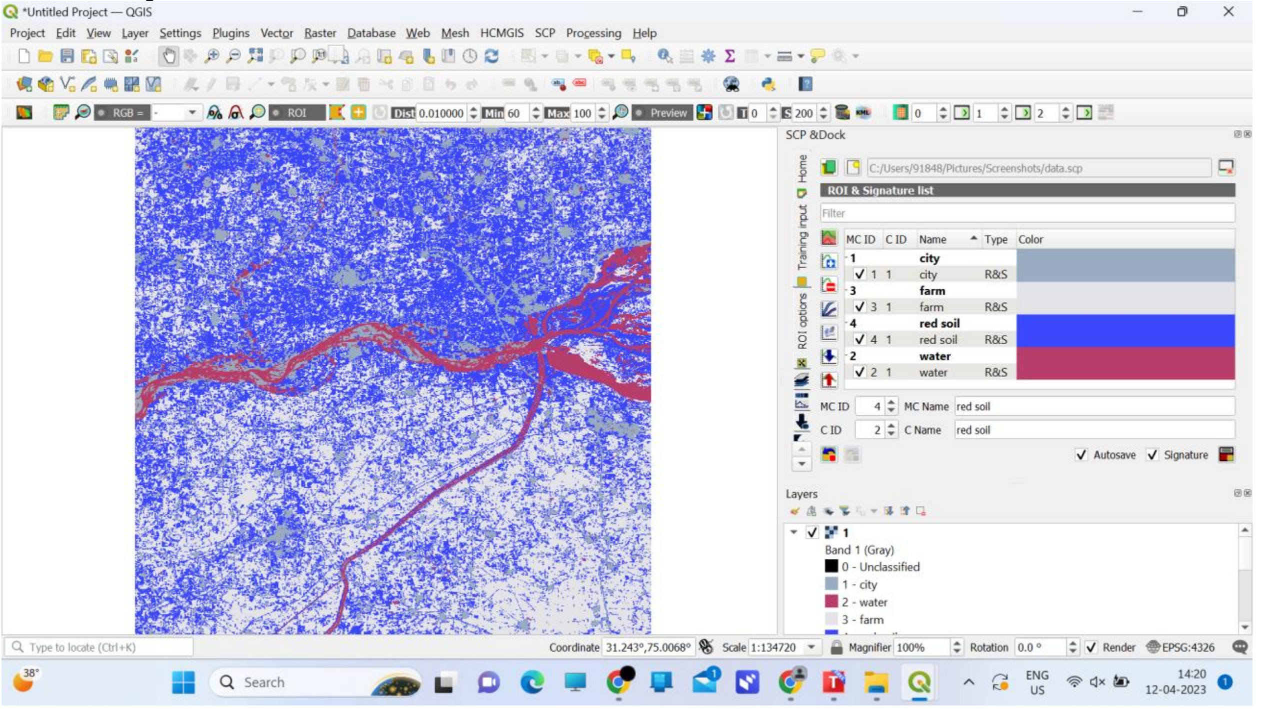


Fig 8.55 Output

- For better visualization, if we change the color of our data

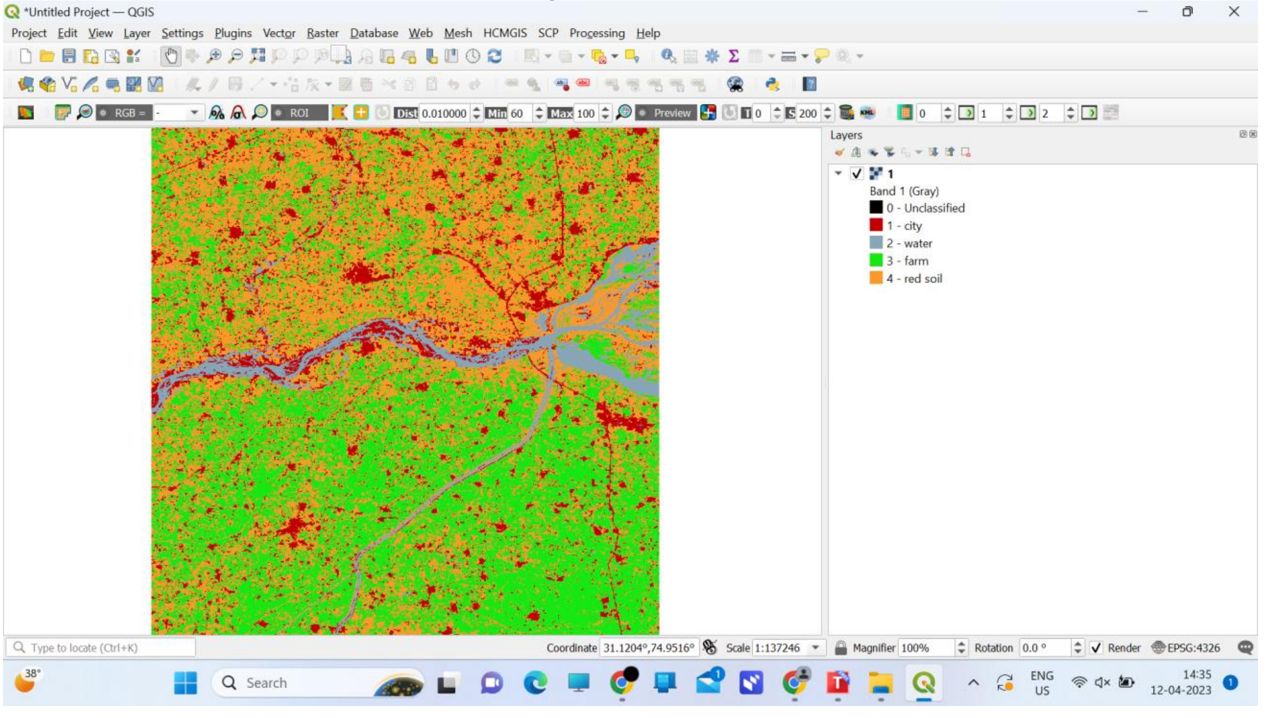


Fig 8.56 Enhanced Output

Classification Report

- This is QGIS generated classification report

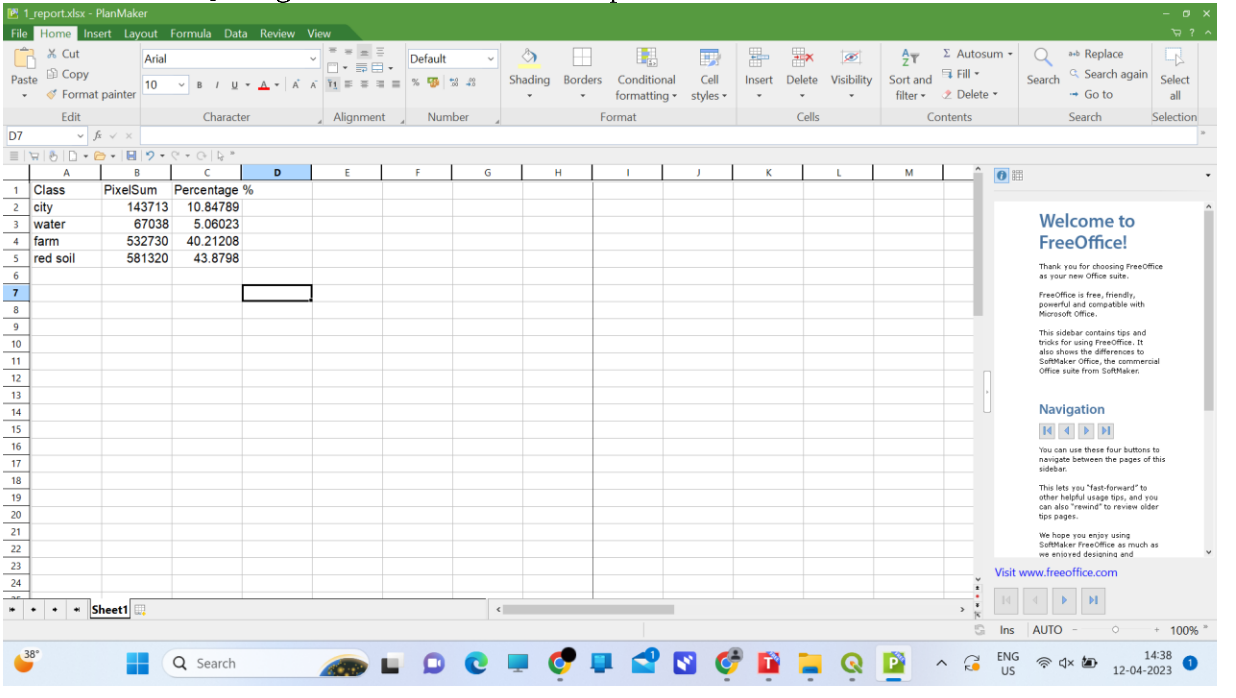


Fig 8.57 Report

9. LIMITATION AND FUTURE ENHANCEMENT

9.1. Limitation

- We are currently collecting very limited amount of data from users due to limitations of identity checking database of government like Aadhar-card etc.

- As Government should be the one In-charge of this web-portal this portal should have a better security of user's data.

- The Internet must be on the whole time while using the web application.

9.2 Future Enhancements

- We will try to collect all types of data needed of users to Authenticate the users identity and reduce redundancy .

- Improving the efficiency of the website.

- Keep on changing the UI-Design to make the website look more attractive.

- Implementing the dark mode feature for the users who are interested to show our website in dark mode.

- Implementing the functionality in which all the users must be verified by developers so any misbehaviour will not be happened

10. CONCLUSION AND DISCUSSION

10.1 Conclusion

According to us, this project gave all of us the confidence to believe in ourselves and a great experience of how to work as a team. It also boosted our requirement gathering, system analysis, designing aspects, technical coding as well as time management skills.

Also, we learned how to work together as a team & collaborate for making ends meet for our web app. We also got an insight into how we would have to work in future at job or startup & how we must contribute for the good of the entity we are working.

10.2 Discussion

10.2.1 Self-Analysis of Project Viabilities

According to us, this project is absolutely a good start for gaining hands-on experience on projects. It is useful if it is managed according to the goal for which it is made.

10.2.2 Problems Encountered and Possible Solutions

There are so many problems encountered during this project.

- technical problems like maintenance of website, data redundancy
- untechnical aspects like effective teamwork, better communication of our team members' individual ideas & combine them for betterment of the website.

10.2.3 Summary of Project Work

It is a great achievement to successfully complete the project. The prior knowledge of software engineering has helped immensely in overcoming the various roadblocks. We have done work with pre-planned scheduling related with time constraints and weekly progress in project development. Also we received guidance from **Prof. (Dr.) Vipul K. Dabhi and Prof. (Dr.) Harshad B. Prajapati** at all stages of our project which helped in overall betterment of the project.

All in all, it was a beautiful experience which taught us many things needed to succeed further in life during our jobs.

11. REFERENCES

- MongoDB: https://www.mongodb.com/docs/
- Stackoverflow: https://stackoverflow.com
- Geeksforgeeks: https://www.geeksforgeeks.org/
- Youtube: https://www.youtube.com/
- W3 Schools: https://www.w3schools.com/
- Cloudinary: https://cloudinary.com/documentation
- React JS: https://reactjs.org/docs/getting-started.html
- Npm: https://www.npmjs.com/
- Icons8: https://icons8.com/
- Font-awesome: https://fontawesome.com/icons
- Materialui: https://mui.com/components/material-icons/